Summary Report of Tongji- JTEKT Joint Lab Project

Online personalized preference learning system
- LCC trajectory in curve

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Abstract

Advanced Driver Assistance Systems (ADAS) are expected to improve driving safety and comfort, but their effectiveness depends on matching the system's behavior with the driver's preferences. Existing research mostly focuses on preference learning in static environments. However, in complex curved scenarios with dynamic interference from adjacent vehicles, driver preferences are more diverse and difficult to quantify. This project takes the Lane Centering Control (LCC) system in dynamic dual-lane curved scenarios as the research object and proposes an online personalized preference adaptation method based on reinforcement learning.

Firstly, to address the challenge of quantifying risk and efficiency in dynamic environments, a dual-lane curved scenario with adjacent vehicle interference is designed. Subjective Risk Perception (RP) indicators are designed. A trajectory planning and tracking model based on Dynamic Programming (DP) is established to quantify safety and efficiency utility in dynamic scenarios. Secondly, to obtain driver-preferred trajectories under complex dynamic constraints, an initial utility model is built based on Maximum Entropy Inverse Reinforcement Learning (IRL). The reward function weights derived from expert trajectories are used to guide the real-time planning algorithm to generate diverse trajectories in the presence of dynamic obstacles.

Finally, an Online Adaptive System (OAS) based on Twin Delayed Deep Deterministic Policy Gradient (TD3) is proposed. The system innovatively introduces a driver intervention mechanism into the loss function, allowing the agent to update policy network parameters directly through driver takeover behaviors, thereby achieving online learning of personalized preferences. Driver-in-the-loop experiments show that the system can converge quickly within an average intervention time of 10 seconds, with an algorithm update time of only 0.75 seconds. Compared with the generalized policy, the personalized policy significantly improves driver satisfaction and safety-efficiency metrics in dynamic curved scenarios.

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1 Introduction

1.1 Research Background

Advanced Driver Assistance Systems (ADAS), such as Adaptive Cruise Control (ACC), Lane Keeping Assistance (LKA), and Lane Centering Control (LCC), have become increasingly common in modern vehicles. ADAS is expected to improve driving safety and comfort, but its effectiveness relies heavily on driver acceptance, which requires the system to match drivers' preferences[1-2]. However, drivers' preferences differ significantly due to personality traits and driving experience. Furthermore, preferences can vary within individual drivers depending on their state and the driving situation.

In static or simple scenarios, preference mainly manifests as comfort. However, in dynamic scenarios involving interference from adjacent vehicles, preference relates more to risk perception and safety margins. Research indicates that different drivers have significantly different tolerances for dynamic risks[3]. Therefore, to achieve harmonious human-machine co-driving in complex dynamic traffic, it is essential to develop personalized ADAS capable of adapting to individual risk perceptions and driving styles[4].

1.2 Research Status

Generally, personalization methods are categorized into explicit and implicit personalization[5]. Explicit personalization requires drivers to manually choose system settings (e.g., time gaps in ACC). However, this is often limited by pre-defined choices, and drivers may find it difficult to understand the engineering implications behind the settings.

Implicit personalization, which derives a personalized driver preference model from collected driving data, is a promising solution. Early implicit methods focused

on driving style identification, classifying drivers into discrete categories like "aggressive" or "conservative"[6]. However, this approach lacks the granularity to meet individual needs. Another widely researched method is Imitation Learning, which trains the system to mimic the driver's natural behavior[7]. However, studies have shown that the assumption "drivers prefer the vehicle to drive like themselves" is not always true. Especially in high-risk dynamic scenarios, drivers often expect the system to drive more safely and robustly than they do themselves[8-9].

Recently, Reinforcement Learning (RL) has shown great potential in autonomous driving decision-making and control. To align RL agents with human preferences, Inverse Reinforcement Learning (IRL) is used to recover reward functions from expert demonstrations[10]. However, most existing IRL methods are trained offline and struggle to adapt to specific users in real-time. Furthermore, in human-machine co-driving, driver "intervention" (taking over control) is the most direct negative feedback signal. How to utilize these sparse but high-value intervention data, combined with Deep Reinforcement Learning (e.g., TD3 algorithm), to achieve rapid online evolution of the system remains a challenging frontier in personalized ADAS research[11-12].

In conclusion, several problems need to be addressed when designing personalized ADAS for dynamic scenarios:

1. How to quantify subjective risk perception and objective safety/efficiency in dynamic environments (e.g., curved roads with adjacent vehicles)?
2. How to build a utility model that reflects the driver's underlying intent rather than simple trajectory imitation?
3. How to utilize driver interventions as feedback to update the control policy online, quickly, and accurately using reinforcement learning algorithms?

1.3 Research Contents

Lane centering control (LCC) system could control the vehicle to drive within the lane and reduce the driving workload a lot. Few researches on LCC trajectory

personalization. This research takes LCC trajectory in curve as the research object. Based on the research review, the research contents are:

1. Validate if there is necessity to adapt to driver to driver preference or one trajectory that most drivers prefer. And validate whether the assumption, that drivers prefer the LCC to drive like themselves is true or not.
2. Find out the better query form to learn driver preference and build driver preference model based on the query answer.
3. Design an online personalized preference learning system, which could learn preference as quick and accurate as possible.

2 Quantifying risk and efficiency in the dynamic driving environment

Following the analysis of research status and the definition of project objectives in Chapter 1, the primary challenge for realizing a personalized LCC system lies in translating the abstract concept of "driver safety perception" into a computable mathematical model. Therefore, before delving into the trajectory planning and control algorithms, this chapter focuses on quantifying driver risk perception in the dynamic driving environment. Specifically, this chapter establishes a novel risk metric system tailored for dynamic curves involving adjacent vehicles, conducts subjective evaluation experiments to map objective indicators to subjective perception using Stevens' Power Law, and finally verifies the rationality of the proposed model through natural driving data. The derived quantitative risk perception model will serve as the crucial safety constraint for the trajectory planning module detailed in the subsequent.

2.1 Design of Risk Perception Metric System in Dynamic Scenarios

2.1.1 Classification and Influencing Factors of Subjective Risk Perception

Previous studies proposed that drivers' risk perception comprises two components: latent risk and overt risk [18-19]. Theoretically, latent risk refers to the risk that arises from external environmental disturbances, system failures, or operational uncertainties when a vehicle is driving safely. For example, in the top left scenario of Figure 2.1, the vehicle is performing a normal turning maneuver, with its front end pointed toward the road boundary. However, as the vehicle gets closer to the lane boundary and its speed increases, the risk of collision with the boundary due to

operational uncertainties becomes more significant. Although the likelihood of such an accident is low, it nonetheless creates a persistent sense of latent risk for the driver.

In contrast, overt risk refers to the collision risk that exists when a vehicle is already in a hazardous situation, such as having deviated from the lane and continuing to travel in its current state. For instance, in the bottom right scenario of Figure 2.1, the vehicle is gradually drifting out of its lane. The shorter the travel path and the higher the speed, the greater the risk of the vehicle colliding with the lane boundary.

Since the driver's source of information is primarily the image in the forward field of view, objects in the driver's field of view that are capable of generating risk are the main influences on the driver's subjective risk perception. In many cases in the dynamic cornering scenario with an adjacent vehicle, the driver's forward visual field is usually occupied by two objects: the static lane boundaries and the dynamically moving adjacent vehicle. The relative positions and motions of these objects and ego vehicle need to be analyzed critically.

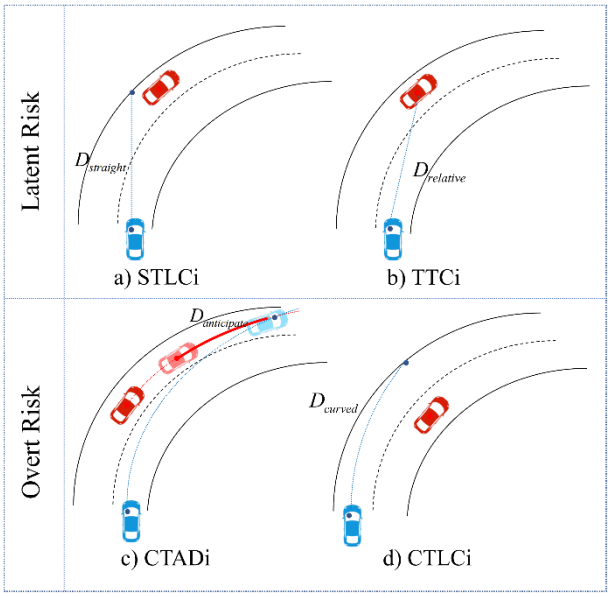


Figure 2.1 Schematic diagram of objective risk indicators.

2.1.2 Design of objective risk perception indicators in dynamic scenarios

To objectively describe the influencing factors of drivers' subjective risk perception and facilitate quantitative modeling, it is imperative to gain a profound understanding of the mechanism governing drivers' risk perception. Consequently, in this study, there is a need to reconfigure objective risk indicators for the examined scenarios based on existing quantitative models of drivers' subjective risk. In previous quantitative studies of driver risk perception, the objective risk indicators are designed as the inverse of the time distance. This is because the reciprocal of the time distance can accurately represent the optical flow information of objects in the driver's vision [16][20], and the driver uses this information to obtain the current vehicle state and predict the future vehicle state. Therefore, it is reasonable and interpretable to design an objective risk indicator by the inverse of the time distance.

this study, based on previous research, proposed four objective risk indicators for the dynamic cornering scenario with adjacent vehicle on a two-lane road. These risk indicators aimed to establish a new risk perception measurement system. As shown in Figure 2.1, STLCi and TTCi indicate latent risk while CTADi and CTLCi indicate overt risk when cornering. The blue and red vehicle represent the ego and adjacent vehicle, respectively. In addition, the light blue and red vehicle represent the anticipated position of them. The blue solid point is the driver's position of the ego vehicle, while the solid red point is the center of gravity of the adjacent vehicle. The blue and red dashed curve are the anticipated trajectory of the ego and adjacent vehicle. The following is the specific concept of these indicators

a) Inverse of straight time to line crossing (STLCi): it represents the inverse of the time distance when the ego vehicle maintains its current speed and the steering wheel angle is $0°$, indicating the time until crossing the lane boundary. Specifically, STLCi calculates the risk margin when the vehicle maintains a constant heading angle and approaches the lane boundary, characterizing the risk posed by the lane boundary

to the driver. STLC received limited attention in lateral deviation studies because many studies in the field of longitudinal tracking control considered STLC as TLC, and did not distinguish between STLC and CTLC. However, the fact is that STLC and CTLC play very different roles in influencing driver risk perception. The results of the study [21] showed that drivers anticipated the STLC of the vehicle's current state in the road environment and adjusted the way they entered a corner to obtain a larger STLC, thus reducing the risk during cornering.

b) Inverse of time to collision (TTCi): it represents the inverse of the time until a collision, calculated by dividing the relative distance between the ego vehicle and the adjacent vehicle by their relative velocity. TTCi describes the risk margin of colliding with the vehicle in front when there is a vehicle ahead. TTC was extensively studied in longitudinal control, particularly in highway straight-line following scenarios. Related studies[17][22][23] indicated that drivers perceive the proximity to the vehicle in front of them based on the TTCi, which created the latent perceived risk to the driver during steady driving. In stable driving conditions, it introduces a latent perception of risk to the driver. Given the curved scenario studied in this study and the different lanes of the ego and adjacent vehicle, the relative velocity of ego and adjacent vehicle to relative distance of them ratio is used to calculate TTCi, aligning with the inherent principles of the driver's visual perception according to the τ theory[16].

c) Inverse of curved time to anticipated distance (CTADi): it is inverse of the time distance calculated by the relative velocity of the two vehicles along the tangent direction of the adjacent vehicle's anticipated trajectory divided by the distance between the two vehicles along the adjacent vehicle's anticipated trajectory at this specific time, while the anticipated trajectories of the ego vehicle and the adjacent vehicle are traversing at their current speed and heading angle. CTADi represents the probability of a future collision between the two vehicles while maintaining their current states. It quantifies the risk margin of the two vehicles colliding in the future from the perspective of time headway. During curve driving, drivers estimate the future states of both vehicles based on the optic flow information in their current field

of view. The smaller the probability of a future collision between the two vehicles, the smaller the perceived risk for the driver.

d) Inverse of curved Time to Line Crossing (CTLCi): it represents the inverse of the time until crossing the lane boundary when the ego vehicle maintains a constant speed and steering wheel angle. CTLCi, from the viewpoint of time headway, quantifies the risk margin of the ego vehicle touching the lane boundary while maintaining its current state. Some studies have been conducted on CTLC in lane departure scenarios, and it has found widespread application in lane departure warning systems. E. R. Boer[24] indicated that the driver's acceptance of curve behavior must meet a threshold for CTLC, and theoretical analysis results demonstrated that CTLC provided a strong basis for explaining the driver's choices in lateral position and speed range during curve navigation, emphasizing the significant role of CTLC in the driver's perception of external risks.

The calculation formulas are provided in Equations (2.1)~(2.4).

$$STLCi = \frac{1}{STLC} = \frac{V_{ego}}{D_{straight}} \tag{2.1}$$

$$TTCi = \frac{1}{TTC} = \frac{V_{relative}}{D_{relative}} \tag{2.2}$$

$$CTADi = \frac{1}{CTAD} = \frac{V_{c-relative}}{D_{anticipate}} \tag{2.3}$$

$$CTLCi = \frac{1}{CTLC} = \frac{V_{ego}}{D_{curved}} \tag{2.4}$$

Where V_{ego} is the velocity of the ego vehicle, $V_{relative}$ signifies the relative velocity between the two vehicles, $V_{c-relative}$ is the relative velocity of the two vehicles along the tangent direction of the adjacent vehicle's anticipated trajectory, $D_{straight}$ is the straight distance when the ego vehicle maintains its current speed and the steering wheel turning angle is $0°$ until it crosses the boundary of the lane. $D_{relative}$ is the relative distance between the ego vehicle and the adjacent vehicle.

$D_{anticipate}$ is the curved distance between the ego vehicle and the adjacent vehicle when both are maintaining their current speed and heading angle. This occurs when the anticipated trajectory of the ego vehicle intersects with the anticipated trajectory of the adjacent vehicle, and the curved distance is measured along the predicted trajectory of the adjacent vehicle. D_{curved} is the curved distance when the ego vehicle is traveling while maintaining its current speed and steering wheel angle until it crosses the lane boundary.

Based on the classification and definition of subjective risk perception by the driver, along with the definition of the four designed objective risk indicators, STLCi and TTCi describe the latent discomfort due to the relative positions of the ego vehicle to the lane and adjacent vehicle when no vehicle deviation occurs, serving to characterize latent risk. On the other hand, CTADi and CTLCi describe the significant discomfort to the driver when the ego vehicle deviates from its lane, anticipating a collision with the lane or adjacent vehicle, used to characterize overt risks. To validate these assumptions, subjective evaluation experiments are designed for stable cornering and vehicle departure scenarios, aiming to establish a quantified model for driver risk perception.

2.2 Design of Subjective evaluation experimental

2.2.1 Magnitude Estimation method

In order to model human subjective perception and objective stimulus, the experiment utilized the magnitude estimation (ME) method [25], employing paired comparisons to directly obtain the magnitude of participants' expressed subjective perception of risk. Subsequently, the relationship between the stimulus intensity and the magnitude estimation was established using Stevens' power law[26], which could be calculated by Equation (2.5).

$$\Psi(I) = kI^{a} \tag{2.5}$$

Where k is a proportionality constant that depends on the units used, a is an exponent that depends on the type of stimulus or sensory modality, $\Psi(I)$ is the magnitude of the sensation evoked by the stimulus, specifically referring to the magnitude of estimation in this study, and I is the intensity of the physical stimulus.

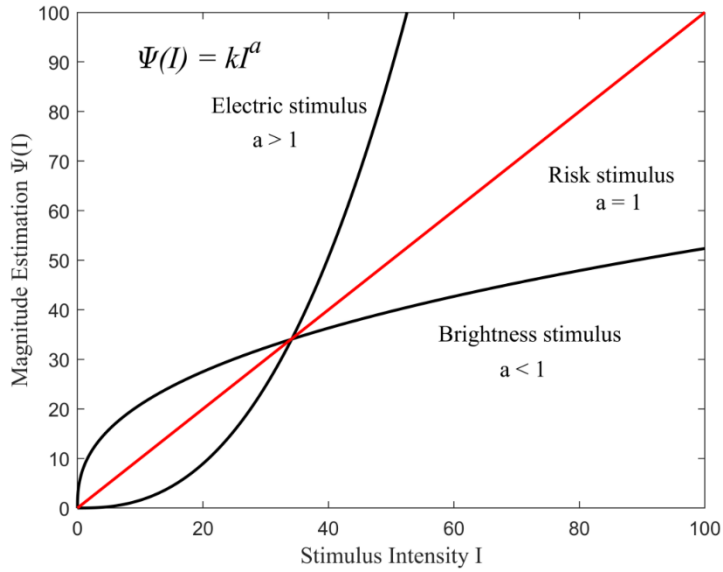


Figure 2.2 Relationship between the stimulus intensity and the magnitude estimation of Stevens' power law

Steven's power law[27], proposed by Steven in 1957, is a fundamental concept in psychophysics that relates the subjective impression of a stimulus to its objective magnitude. Xie et al. introduced a visual model based on combining and extending existing models to successfully predict results related to Steven's power law response. Lee et al. investigated the relationship between Steven's power law and Bartleson-Breneman's brightness function to derive a brightness function for optimal display systems. Kamadia et al. evaluated aroma intensity of various odorants and found that they followed Steven's Power Law, demonstrating the applicability of this law beyond visual perception[28]. Choubey presented a CMOS pixel capable of producing a wide dynamic range response similar to Steven's power law of brightness, highlighting the versatility of this law in technological applications[29]. Furthermore, Baldwin et al. used Steven's Power Law to compare perceived urgency across auditory, visual, and tactile modalities, showcasing the law's relevance in multimodal perception studies[30]. Fakhoury et al. explored the impact of visual cues on haptic

compliance discrimination using a pseudo-haptic robotic system, indicating the potential for Steven's power law to inform research on sensory integration[31]. Overall, Steven's power law has been a valuable tool in understanding perception and response to stimuli across various sensory modalities, from visual brightness to auditory and olfactory stimuli. Its application extends beyond traditional psychophysical studies to technological advancements and multimodal perception research.

Figure 2.2 showed the example of Stevens' power law for different types of stimuli. For features like brightness, the exponent a of the human perception characteristic power function is less than 1, while for electrical stimulus, a is greater than 1. Specific research quantified the risk perception of drivers, and experimental results indicated that the driver's perception of the stimulus intensity of risk is linearly related to the objective stimulus intensity of risk itself (a=1)[23]. Based on this conclusion, this study attempted to construct a complete risk expression. Specifically, the stimulus intensity represents the vehicle's state in the curve, calculated through the previously mentioned objective risk stimulus indicators STLCi, TTCi, CTADi, and CTLCi. In this study, it is assumed that the driver risk perception model is linear, a reasonable assumption is formulated as shown in Equation (2.6):

$$RP_s = \left(kp_1, kp_2, kp_3, kp_4\right) \cdot \begin{pmatrix} STLCi \\ TTCi \\ CTADi \\ CTLCi \end{pmatrix} \tag{2.6}$$

Where RP_s is the driver's subjective risk perception, and kp_1, kp_2, kp_3, kp_4 are constants representing the proportion of each indicator in the total risk stimulus intensity. Subsequently, the subjective evaluation experiment was designed to determine the proportion of each indicator in the composition of risk perception, in order to obtain a comprehensive understanding of the driver risk perception characteristics.

2.2.2 Experimental equipment and scenarios

All the experiments in this study were conducted on a fixed-base simulator. As Figure 2.3 shown, the visual display is based on a high-resolution curved screen (3840×1080), which can cover a vertical viewing angle of about $40°$ and a horizontal viewing angle range of $100°$. Specifically, the experimental scenarios were modeled and designed using a combination of Carla and Roadrunner, where Carla is used to display and control the vehicle motion, and Roadrunner is used to build the map of the test scenario.

Figure 2.3 Fixed-base driving simulator

In addition, the driver's field of view was calibrated during the pre-scene building phase to ensure the realism of the simulator display. A real Model 3 was first parked in the right lane of a two-lane road in a closed environment. Photographs of the driver's field of view were then taken from the driver's eye position. Finally, the driver's field of view in the Carla simulator was calibrated based on the real driver's field of view of a vehicle driving on a dual carriageway. As shown in Figure 2.4, by comparing the positional relationship between the road boundary line in front and the lower edge of the front windscreen of the vehicle, it can be seen that the adjusted scene is very similar to the real field of view. Based on this, the position of the driver's field of view in each scene in the experiment is the fixed position to ensure the realism of the scene display.

a) View of real road scenario

b) View of Carla simulation road scenario

Figure 2.4 Calibration of the driver's view

The real-world road scenario selected for this study is a two-lane highway ramp. The road characteristics primarily considered include curve radius, lane width, and speed range. Based on the recommendations for ramp design speed, curve radius, and lane width from Table 4 and 6 of standard[32], the study selected a test speed range of 35 km/h to 65 km/h, with a lane width of 4 meters and a curve radius of 120 meters. To simplify the dynamic experimental scenario, the experiment involved one dynamic adjacent vehicle and one ego vehicle. Given the double-lane setting, there were two scenarios: the ego vehicle in the inner lane with the adjacent vehicle in the outer lane, and the ego vehicle in the outer lane with the adjacent vehicle in the inner lane. As shown in Figure 2.5, the simulation effect of the driver's perspective when the ego vehicle in the inner lane and the ego vehicle in the outer lane.

a) Ego vehicle in the inner lane and adjacent vehicle in the outer lane.

b) Ego vehicle in the outer lane and adjacent vehicle in the inner lane.

Figure 2.5 Front view of the driver in the experimental scenario

2.2.3 Experiment procedure

The experiments were designed to both stable cornering conditions and lane departure conditions respectively. A total of 16 drivers participated in this study, including 5 professional drivers and 11 non-professional drivers. As shown in Table 2.1, professional drivers had an average driving experience of approximately 12 years, possessing extensive driving experience and proficient driving skills. Non-professional drivers, on the other hand, had an average driving experience of around 4 years, with relatively less driving experience.

Table 2.1 Participant Information of Subjective Evaluation Experiment

No.	Gender	Driving age	Driving frequency	Career
1	Male	19	Often	Professional driver
2	Male	13	Often	Professional driver
3	Male	13	Often	Professional driver
4	Male	7	Often	Professional driver
5	Male	7	Often	Professional driver
6	Male	6	Sometimes	Graduate student
7	Male	6	Sometimes	Graduate student
8	Male	5	Sometimes	Graduate student
9	Female	5	Sometimes	Graduate student
10	Male	5	Sometimes	Graduate student
11	Male	4	Often	Graduate student
12	Male	3	Sometimes	Graduate student

13	Female	3	Sometimes	Graduate student
14	Male	2	General	Undergraduate
15	Male	1	Sometimes	Graduate student
16	Male	1	Sometimes	Graduate student

Experiment 1: Quantification of driver's subjective risk perception in stable cornering conditions. This experiment aimed to explore the relationship between the driver's latent risk perception and the objective risk indicators, STLCi and TTCi.

Experiment 2: Quantification of driver's subjective risk perception in lane departure conditions. This experiment aimed to explore the relationship between the driver's overt risk perception and the objective risk indicators, CTLCi and CTADi.

1） The screen displays the standard stimulus scenario (Experiment 1: 12s; Experiment 2: 10s)

2） Screen pauses for 1s

3） The screen displays the comparative stimulus scenario for the same duration as in step 1

4） Driver reports how many times the perceived risk of the comparative stimulus scenario has increased verbally compared to the standard stimulus scenario

Figure 2.6 Experimental steps of the magnitude estimation method

The specific experimental steps are illustrated in Figure 2.6:

1. Drivers observe standard stimulus scenarios on the computer screen for a certain duration, where each row in Table 2.3 and Table 2.4 represents a stimulus scenario, and the standard stimulus scenarios are determined according to Table 2.2 (Experiment 1: 12s; Experiment 2: 10s).
2. The screen freezes for 1s.
3. Drivers observe comparison stimulus scenarios on the computer screen, with the same duration as in step 1.

4. Drivers verbally report how much their risk perception has increased compared to the standard stimulus scene, and the value is recorded in the corresponding position in Table 2.2. Then, return to step 1.

Table 2.2 Combinations used in subjective evaluation experiment

		Comparative stimulus						
		Scenario1	Scenario2	Scenario3	Scenario4	Scenario5	Scenario6	Scenario7
Standard stimulus	Scenario1		X	X				
	Scenario2	X		X				
	Scenario3		X		X			
	Scenario4			X		X		
	Scenario5				X		X	
	Scenario6					X		X
	Scenario7					X	X	

Note: Before each formal experiment, drivers were asked to conduct a pre-evaluation to establish a stable evaluation criterion for assessing the magnitude of scenario risk. Based on the ME method, formal experiments were performed based on Table 2.2. Since the risk is continuously changing during vehicle movement, to allow drivers to more accurately assess risk perception in different situations, this study adopted the concept of the "peak-end effect" from literature[33]. The end of the entire test scenario was defined as the driver's evaluation moment. In other words, the driver's evaluation of the whole experience is primarily based on the stimulus peak period during that period and the experience at the end. In the experimental scenario, drivers felt the strongest stimulus at the final moment. Therefore, the end of the test scenario was chosen as the evaluation moment to reflect the driver's perceived experience better. In addition, the study aims to develop an average risk quantification model. Therefore, the subjective risk value for each scenario is not the score of a specific driver but the average of all drivers' subjective scores for that scenario. Finally, each driver in the subjective evaluation experiment will obtain 82 sets of raw data, and the 16 drivers will obtain a total of 1344 sets of experimental data.

2.3 Experiment 1: Quantification of Driver Risk Perception in Stable Cornering Condition

2.3.1 Experimental parameter configuration

This experiment aimed to quantify the driver risk perception in stable cornering conditions. Stable cornering conditions refer to both the ego vehicle and the adjacent vehicle maintaining a fixed speed and lateral offset. In such conditions, there is no deviation of vehicles from the lane, and there is no overt risk, meaning that the values of CTADi and CTLCi are equal to 0. Only latent risk needs to be considered, so it is necessary to distribute the parameters STLCi and TTCi to design scenarios with different objective risk stimulus intensities, allowing drivers to perceive differences between various scenarios.

Based on the definitions of STLCi and TTCi and the design of the experimental scenario, different risk scenarios were designed by changing the speed of the ego vehicle while keeping the speed of the adjacent vehicle constant. The details are shown in Table 2.3, with seven scenarios of different risk stimuli distributed for each working condition, adding up to a total of 14 base scenarios. The experiment is then conducted according to the paired comparison groups in Table 2.2.

Table 2.3 Distribution of objective risk indicators for experiment 1

No.	Ego vehicle in the inner lane			Ego vehicle in the outer lane		
	$V_{ego}(km/h)$	$STLCi(s^{-1})$	$TTCi(s^{-1})$	$V_{ego}(km/h)$	$STLCi(s^{-1})$	$TTCi(s^{-1})$
1	35	0.278	0.030	35	0.552	0.027
2	40	0.318	0.159	40	0.630	0.086
3	45	0.358	0.288	45	0.708	0.199
4	50	0.397	0.418	50	0.786	0.312
5	55	0.437	0.549	55	0.864	0.420
6	60	0.476	0.680	60	0.942	0.533
7	65	0.516	0.813	65	1.019	0.654

2.3.2 Results of drivers' subjective evaluation in Stable cornering condition

The subjective risk data for drivers was obtained through comparative experiments. The initial data consisted of risk ratios between different scenarios, rather than the absolute risk values for each scenario. To derive the absolute risk value for each scenario, one of the 7 experimental scenarios was first selected as the standard scenario, and its subjective risk value was set to 1. For instance, as shown in Figure 2.7, Scenario 4 was chosen as the standard scenario with a subjective risk value of 1. Using the standard scenario and its comparison group as a starting point, the absolute risk values for the remaining 6 scenarios were calculated based on the risk ratios between each pair of scenarios. Theoretically, once the standard scenario is assigned a value of 1, a minimum of 6 comparison groups is required to calculate the absolute risk values for all 7 scenarios.

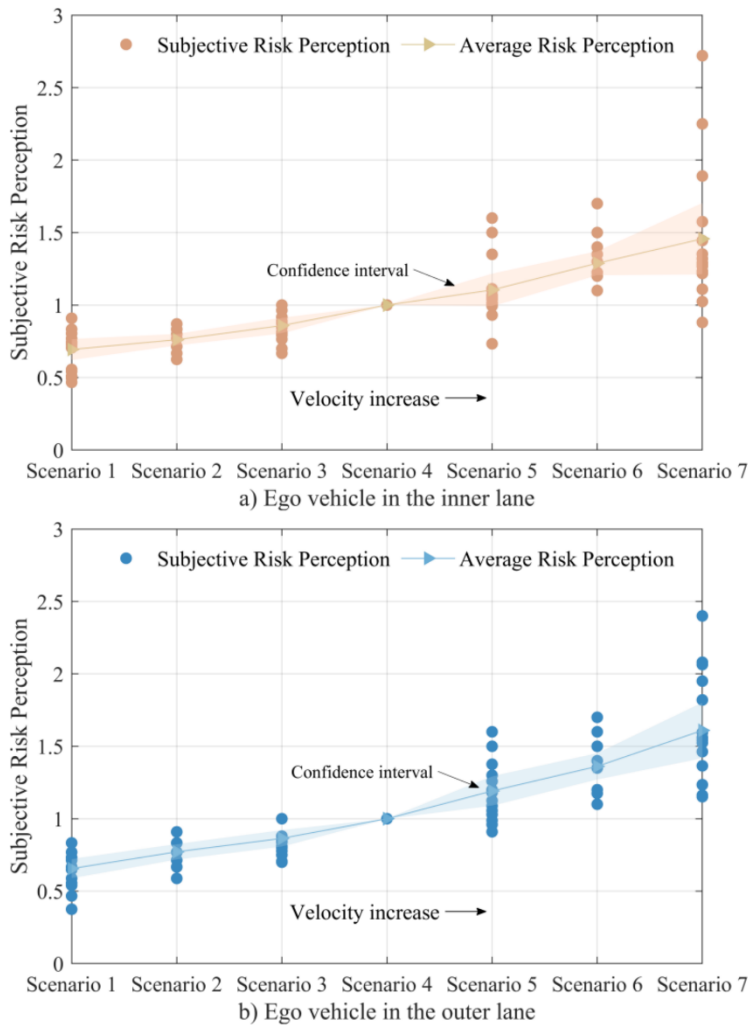


Figure 2.7 Result of the subjective evaluation of drivers in experiment 1

From Figure 2.7, it could be observed that, regardless of whether the ego vehicle was in the inner lane or outer lane, drivers' risk perception increased with the increase in the ego vehicle's speed in the scenario. Moreover, as evident from the light-colored band-shaped confidence interval in the figure, the 95% confidence interval indicated that there was a 95% probability for participants' assessment results to fall within this interval. Additionally, the trend of the interval was consistent with the trend of the mean, suggesting that the overall distribution of the data had a low level of dispersion.

2.3.3 Model of Driver Perception of latent Risk

Based on the ME method and the fact that the driver risk perception in stable cornering conditions mainly consisted of latent risk, which was characterized by STLCi and TTCi, Equation (2.6) can be transformed into:

$$RP_{latent} = kp_1 \times STLCi + kp_2 \times TTCi \tag{2.7}$$

Where RP_{latent} represents the objective latent risk stimulus, kp_1 and kp_2 are weighting coefficients, representing the weights of STLCi and TTCi in the overall objective latent risk stimulus.

Therefore, utilizing Equation (2.7), the objective risk stimulus for each scenario can be computed. To keep the driver's subjective risk perception and the objective risk stimuli of the scenarios on the same scale, the objective risk stimulus of scenario 4 was also set to 1, and the objective risk stimuli of the remaining scenarios were scaled proportionally. Subsequently, Stevens' power law equation was applied to perform a least squares fitting of the driver's subjective risk perception and the objective risk stimulus of the scenario. The aim was to determine the optimal values of kp_1 and kp_2 that maximize the linear fit.

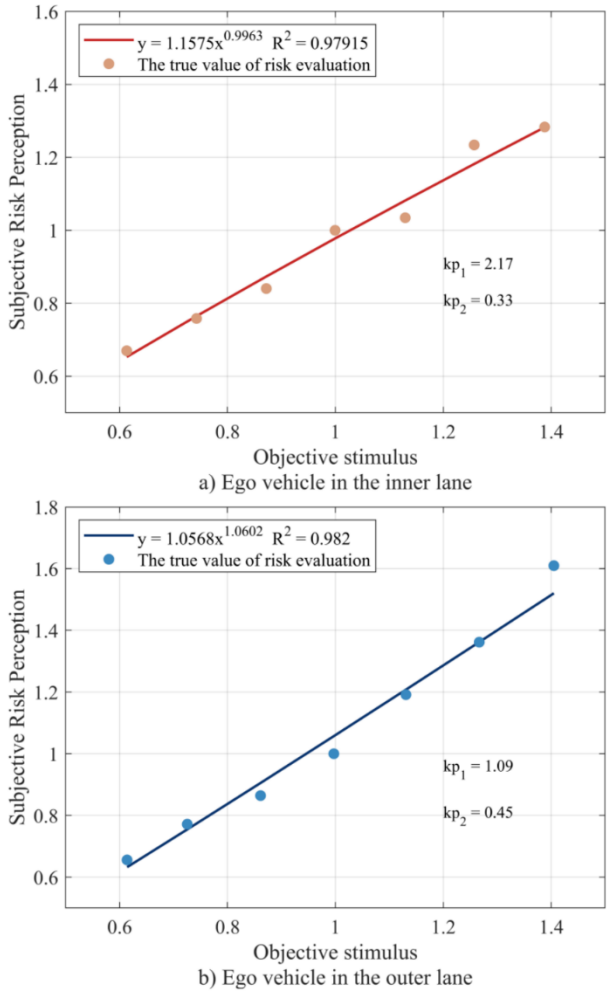


Figure 2.8 Relationship between objective risk stimulus and subjective risk perception in experiment 1

The results are depicted in Figure 2.8. For the scenarios with the ego vehicle in the inner lane, the model achieved the highest fitting accuracy when $kp_1 = 2.17$ and $kp_2 = 0.33$. Similarly, for the scenarios with the ego vehicle in the outer lane, the optimal fitting occured when $kp_1 = 1.09$ and $kp_2 = 0.45$. In both cases, the final results exhibited an approximation to linearity, with correlation coefficients surpassing 0.97. This suggested that in stable cornering conditions, the driver's subjective risk perception is proportional to the objective risk stimulus (RP_{latent}) calculated through a linear combination of STLCi and TTCi. Therefore, the model for the driver's latent risk perception could be expressed using Equations (2.8) and (2.9).

$$RP_{latenti} = 2.17 \times STLCi + 0.33 \times TTCi \tag{2.8}$$

$$RP_{latento} = 1.09 \times STLCi + 0.45 \times TTCi \tag{2.9}$$

where $RP_{latenti}$ and $RP_{latento}$ denote the driver's subjective latent risk perception when the ego vehicle in the inside lane and outside lane, respectively.

2.4 Experiment 2: Quantification of Driver Risk Perception in Lane Departure Condition

2.4.1 Experimental parameter configuration

The lane departure conditions refer to the scenarios where the ego vehicle and the adjacent vehicle maintain the current speed and steering wheel angle, and the ego vehicle gradually deviates towards the outer lane boundary while the adjacent vehicle continues to drive in its centerline. According to the definition of objective risk indicators, when the ego vehicle in the inner lane and the adjacent vehicle in the outer lane, the ego vehicle deviating outward implies the overt risk of collision with the adjacent vehicle. In this condition, the overt risk of a vehicle collision is much greater than the risk of crossing the lane boundary, so the overt risk for this condition was characterized by CTADi. To ensure uniform latent objective risk for each experimental group and distribute CTADi and CTLCi for each scenario, the paper adjusted the curvature of the ego vehicle's trajectory, thus altering the overt risk stimulus for each scenario. As illustrated in Table 2.4, 7 scenarios with different risk stimuli were distributed for each condition, totaling 14 basic scenarios. Subsequently, experiments were conducted based on the paired comparison groups defined in Table 2.2.

Table 2.4 Distribution of objective risk indicators for Experiment 2

	Ego vehicle in the inner lane			Ego vehicle in the outer lane		
	$Curvature(m^{-1})$	$CTADi(s^{-1})$	$CTLCi(s^{-1})$	$Curvature(m^{-1})$	$CTADi(s^{-1})$	$CTLCi(s^{-1})$
1	0.0009	1.086	-	0.0009	-	1.295
2	0.0021	0.936	-	0.0022	-	1.233
3	0.0033	0.809	-	0.0035	-	1.157
4	0.0045	0.694	-	0.0049	-	1.063
5	0.0056	0.568	-	0.0062	-	0.942

6	0.0068	0.447	-	0.0075	-	0.783
7	0.008	0.327	-	0.0088	-	0.508
		$STLCi = 0.535s^{-1}$			$STLCi = 1.368s^{-1}$	
		$TTCi = 1.111s^{-1}$			$TTCi = 0.467s^{-1}$	

2.4.2 Results of drivers' subjective evaluation in Lane Departure condition

As shown in Figure 2.9, scenario 4 was chosen as the standard stimulus scenario, with its subjective risk perception set to 1. The findings revealed that irrespective of the ego vehicle being in the inner or outer lane, drivers' subjective risk perception gradually decreased with an increase of the ego vehicle's trajectory curvature. When considering this alongside Table 2.3, it became evident that as CTADi and CTLCi increased, drivers' subjective risk perception became more pronounced. The light-colored banded confidence intervals in the figure revealed a 95% probability that participants' assessments fell within this interval. Moreover, the interval's trend aligned with the mean, indicating a less discrete overall data distribution.

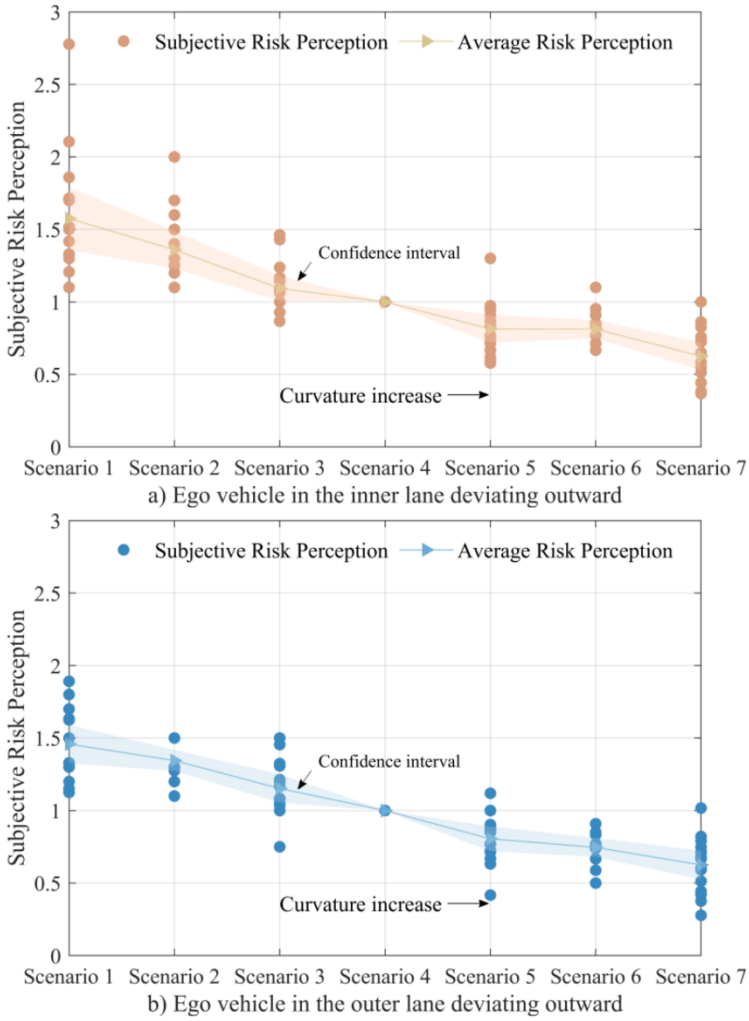


Figure 2.9 Results of the subjective evaluation of drivers in experiment 2

2.4.3 Model of Driver Perception of overt Risk

Since the overall risk stimulus of the lane departure condition is stronger, to prevent the dispersion of the number of subjective risk perceptions of the driver, the standard stimulus scenario was changed to the riskiest scenario 1. Thus, the subjective risk perception and objective risk stimulus of scenario 1 were set to 1, and the values of the rest of the scenarios were scaled equivalently before the model fitting was carried out. As stated earlier, the driver's subjective overt risk perception was only characterized by CTADi or CTLCi. Therefore, the objective overt risk stimulus for each scenario was calculated by Equations (2.10) and (2.11).

$$RP_{overti} = CTADi \tag{2.10}$$

$$RP_{overto} = CTLCi \tag{2.11}$$

Where RP_{overti} represents the objective overt risk stimulus when the ego vehicle in the inner lane, and RP_{overto} represents the objective overt risk stimulus when the ego vehicle in the outer lane.

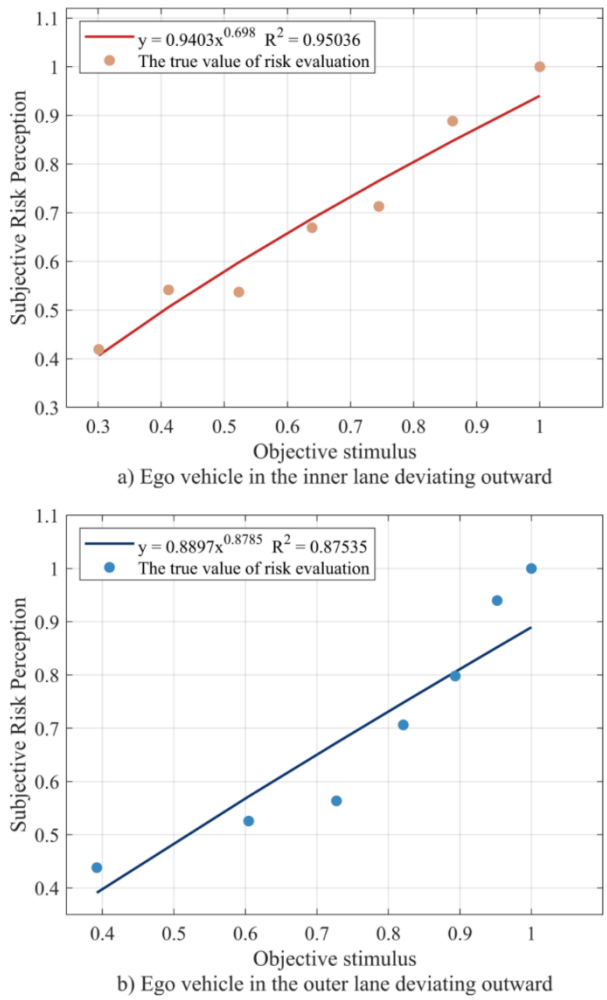


Figure 2.10 Relationship between objective risk stimulus and subjective risk perception without considering the latent risk

Figure 2.10 illustrates the relationship between objective overt risk and drivers' subjective risk perception when latent risk is not considered. Subsequently, a least-squares fit was applied to the Stevens power law equation. The fitted exponent values were found to be 0.698 and 0.8785. The results showed that although there is a positive correlation between drivers' subjective risk perception and objective overt risk stimulus in experiment 2, the fitting results do not align with the assumption that

drivers' subjective risk perception and objective overt risk stimulus follow a linear pattern. The reason for this might be that, in the lane departure scenario, drivers' subjective risk perception is composed of two parts: latent risk and overt risk. Therefore, when calculating objective risk stimulus, both aspects need to be considered simultaneously. The overall objective risk stimulus for each scenario can be computed using Equations (2.12) and (2.13).

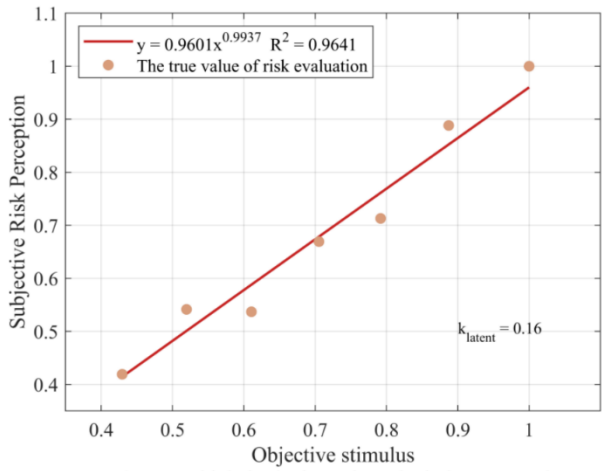
$$RP_{totali} = RP_{overti} + k_{latenti} \times RP_{latenti} \tag{2.12}$$

$$RP_{totalo} = RP_{overto} + k_{latento} \times RP_{latento} \tag{2.13}$$

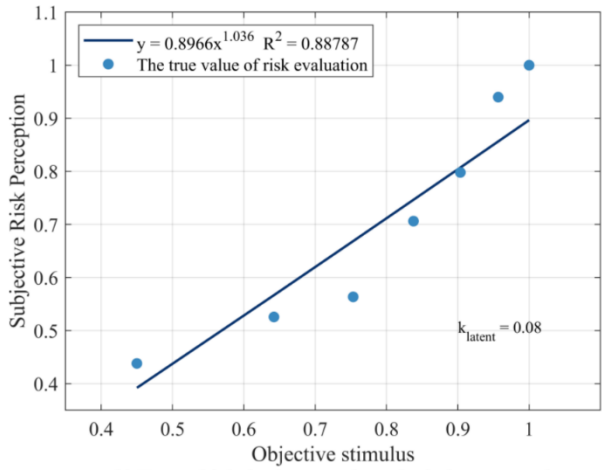
Where RP_{totali} and RP_{totalo} respectively represent the total objective risk stimulus when the ego vehicle in the inner and outer lane. $k_{latenti}$ and $k_{latento}$ respectively represent the proportion of objective latent risk stimulus to the overall objective risk stimulus when the ego vehicle in the inner and outer lane. Similarly, selecting the scenario with the maximum risk, i.e., Scenario 1, as the standard scenario, the objective risk stimulus for the remaining scenarios was calculated based on Equation (2.14).

$$RP_{com} = \frac{RP_{overt|com} + k_{latent} \times RP_{latent|com}}{RP_{overt|std} + k_{latent} \times RP_{latent|std}} \tag{2.14}$$

Where RP_{com} represents the objective risk stimulus size for the comparison stimulus scenario. $RP_{overt|std}$ and $RP_{latent|std}$ represent the apparent and potential risks for the standard stimulus scenario. $RP_{overt|com}$ and $RP_{latent|com}$ represent the apparent and potential risks for the comparison stimulus scenario.



a) Ego vehicle in the inner lane deviating outward



b) Ego vehicle in the outer lane deviating outward

Figure 2.11 Relationship between objective risk stimulus and subjective risk perception considering the latent risk

The results of the final fitting, as illustrated in Figure 2.11, indicate that when the ego vehicle is in the inner lane with $k_{latenti}$ at 0.16, the exponent of the fitted curve is 0.9937. Similarly, when the ego vehicle is in the outer lane with $k_{latento}$ at 0.08, the exponent of the fitted curve is 1.036. These findings suggest that the driver's subjective risk perception is directly proportional to the objective risk stimulus when the latter is calculated using Equations (2.15) and (2.16), allowing the objective risk stimulus to be used as a surrogate for the driver's subjective risk perception.

$$RP_{totali} = RP_{overti} + 0.16 \times RP_{latenti} \tag{2.15}$$

$$RP_{totalo} = RP_{overto} + 0.08 \times RP_{latento} \tag{2.16}$$

2.5 Validation of the risk perception model based on driver-in-the-loop simulation experiment

2.5.1 Design of Natural Driving Experimental

Since drivers are typically in a highly focused state when entering curves and are almost always in a stable cornering condition, this section was designed to conduct a driver-in-the-loop simulation experiment, collecting natural driving data from drivers to validate the quantitative risk perception model by using risk boundary which is able to relect driving behavior for stable cornering conditions[0].

Table 2.5 Participant Information of Natural Driving Experiment

No.	Gender	Driving age	Driving frequency	Career
1	Female	3	Sometimes	Graduate student
2	Male	7	Often	Professional driver
3	Male	19	Often	Professional driver
4	Male	13	Often	Professional driver
5	Male	13	Often	Professional driver
6	Male	12	Often	Professional driver
7	Male	11	Often	Professional driver
8	Male	6	Often	Professional driver
9	Male	6	Often	Professional driver
10	Male	5	Sometimes	Graduate student
11	Male	1	Sometimes	Graduate student
12	Male	6	Sometimes	Graduate student
13	Male	2	General	Undergraduate
14	Female	4	Sometimes	Graduate student
15	Male	6	Sometimes	Graduate student
16	Male	6	Sometimes	Graduate student

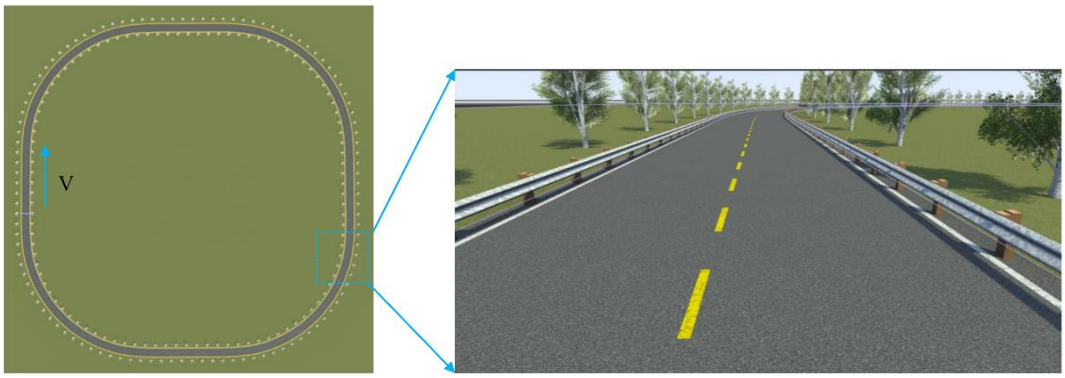


Figure 2.12 Natural driving experiment scenario

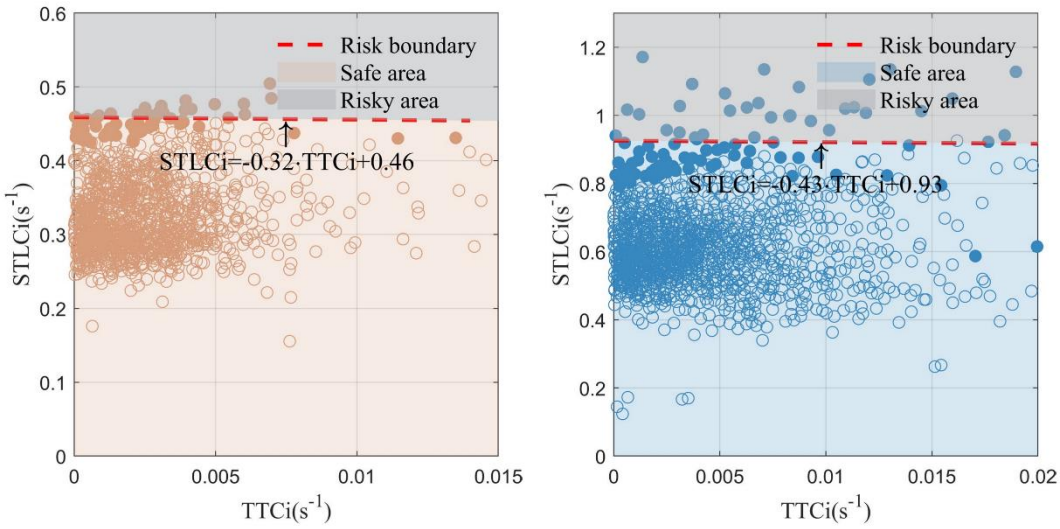
As shown in Figure 2.12, to ensure the experiment could collect a sufficient amount of curved driving data, the entire experimental scenario consisted of four identical curves with a radius of 120m, connected by straight sections where transition curves were employed. The experiment was divided into two cases: the ego vehicle in the inner lane with the adjacent vehicle in the outer lane, and the ego vehicle in the outer lane with the adjacent vehicle in the inner lane. In both cases, drivers were free to control the throttle, brake, and steering wheel while maintaining their lane. To ensure that drivers could perceive the risk from the adjacent vehicle in the forward field of view throughout the entire experiment, the experiment also controlled the speed of the adjacent vehicle, ensuring it remained within the driver's forward field of view.

2.5.2 Validation of driver risk perception model

The experiment enlisted a total of 16 participants, comprising 8 professional drivers and 8 non-professional drivers. The participant details are presented in Table 2.5. The dimension of the natural driving data of each driver in the natural driving experiment is about 4×10^{7}, and the 16 drivers obtained a total of about 1.6×10^{8} sets of data. So the amount of data in this experiment is sufficient.

To validate the subjective risk model presented in the previous sections, evidence from drivers' natural driving behavior is examined. Initially, drivers respond to perceived risks to maintain a safe state. The objective of this study is to determine the risk boundary using Equations (2.8) and (2.9) of the linear risk model proposed earlier.

This boundary represents the acceptable level of risk for the driver. When drivers perceive that the current state exceeds this boundary, they take action to bring the vehicle's state back within the boundary. To derive the risk boundary in the same mathematical form as Equations (2.8) and (2.9), the driving state is determined by the parameters STLCi and TTCi. Since drivers aim to keep the vehicle within the risk boundary, high-risk points are extracted from each interval and a linear fit is performed on these points across all intervals to establish the risk boundary.



a) Ego vehicle in the inner lane b) Ego vehicle in the outer lane

Figure 2.13 Result of the natural driving experiment.

Figure 2.13 illustrates the joint distribution of STLCi and TTCi for each driver's natural driving data when the ego vehicle is in the inner lane and outer lane, respectively. In the figure, circles represent the driver's natural driving data composed of STLCi and TTCi. Solid circles denote risk points, which are points outside the 95th percentile of each transverse fixed interval. The fixed intervals are 0.0025 for the inner lane and 0.0003 for the outer lane, respectively. while hollow circles represent safety points, defined as points other than the risk points. The red dashed line represents the driver's risk boundary. This line is derived by applying a linear regression technique to the solid risk points. Specifically, the risk boundary is determined by fitting a straight line to the mean of the solid circles (risk points) within each fixed interval using the least squares method. The slope and intercept of this

fitted line are specified in the equations below. This method provides a robust estimate of the risk threshold, based on the distribution of the high-risk points.

While individual drivers may vary in their risk perception, this study focuses on establishing a general risk boundary that represents an average risk threshold. This threshold is then compared to the average subjective risk model to examine the alignment between subjective and objective risk perceptions.

The risk boundary Equations (2.17) and (2.18) are listed below.

$$STLCi = -0.32 \times TTCi + 0.46 \tag{2.17}$$

$$STLCi = -0.43 \times TTCi + 0.93 \tag{2.18}$$

The Equations (2.17) and (2.18) can be rearranged as follows:

$$0.47 = 1.03 \times STLCi + 0.33 \times TTCi \tag{2.19}$$

$$0.97 = 1.05 \times STLCi + 0.45 \times TTCi \tag{2.20}$$

Equations (2.19) and (2.20) represent the acceptable risk boundary for drivers navigating the inner and outer lanes, respectively. Mathematically, the left side of these equations denotes the magnitude of acceptable risk, while the right side reflects drivers' perception and quantification of risk under natural driving conditions. By comparing the right-hand sides of Equations (2.19) and (2.20) with those of Equations (2.8) and (2.9), it can be inferred that:

1. A distinction exists between a driver's natural driving behavior and their subjective perception of risk, as evidenced by the differing coefficients in the equations. Similar conclusions have been drawn in prior studies [13-15].
2. Drivers' subjective risk perceptions and driving behaviors exhibit comparable patterns. This conclusion is supported by the observation that risk values are determined through a linear combination of STLCi and TTCi, where the coefficients differ but remain relatively small.

Ultimately, The driver behavior mechanism can be interpreted using Equations (2.19) and (2.20): when the ego vehicle is in the inner lane and the environmental value of ($1.03 \cdot \mathrm{STLCi} + 0.33 \cdot \mathrm{TTCi}$) exceeds 0.47, drivers adjust steering to maintain Equation (2.19) below the risk threshold, reducing perceived risk. This adjustment reduces latent risk, ensuring lane-keeping stability. The same principle applies when the ego vehicle is in the outer lane, and the risk boundary for the outer lane is higher than that for the inner lane, which aligns with real-world conditions.

2.6 Conclusion

In this study, four objective risk perception indicators were devised based on the driver's visual perception characteristics in dynamic curve driving conditions, addressing the lack of objective indicators that mechanically represent the driver risk perception in dynamic scenarios. Furthermore, two sets of subjective evaluation experiments were designed, considering the priori knowledge that the driver's risk comprises latent risk and overt risk. Subsequently, a quantitative model of driver risk perception was proposed by fitting subjective and objective data. Ultimately, the rationality and applicability of the model were verified by natural driving experiment. The summary is as follows:

$$RP_{latenti} = 2.17 \times STLCi + 0.33 \times TTCi \tag{2.21}$$

$$RP_{latento} = 1.09 \times STLCi + 0.45 \times TTCi \tag{2.22}$$

$$RP_{totali} = RP_{overti} + 0.16 \times RP_{latenti} \tag{2.23}$$

$$RP_{totalo} = RP_{overto} + 0.08 \times RP_{latento} \tag{2.24}$$

$$RP_{overti} = CTADi \tag{2.25}$$

$$RP_{overto} = CTLCi \tag{2.26}$$

The quantitative model shows that the driver's perceived risk when the ego vehicle in the inner lane and outer lane can be calculated by two parts, latent risk and overt risk, which are the linear combination of objective risk indicators proposed in this study. What's more, the quantitative model is able to directly calculate the magnitude of the driver risk perception in dynamic curve driving conditions in real time. Therefore, due to its advantages of real-time, simple and interpretable form, the model can be conveniently applied in the calculation and design of safety terms in decision-making and trajectory planning.

3 Trajectory Plan and Track

The drivers' preferred trajectories in curve are diverse. The precondition of personalization is to generate diverse trajectories. This section introduced a trajectory planning and tracking method, which would be used to generate diverse trajectories.

3.1 Trajectory Plan

To ensure the generated trajectories are smooth, continuous, and compliant with vehicle kinematic constraints, the motion planning module utilizes a polynomial-based trajectory generation method operating within the Frenet coordinate system. This approach decouples the planning problem into lateral and longitudinal components, which are subsequently transformed back into the Cartesian coordinate system.

3.1.1 Polynomial Trajectory Planning Algorithm

The trajectory is modeled as a function of planning time t. To satisfy the necessary boundary conditions at both the start and end states, a quintic polynomial is applied for lateral motion $l(t)$, and a quartic polynomial is typically used for longitudinal motion $s(t)$. The trajectory model is defined as:

$$\begin{cases} l(\mathrm{t}) = \mathrm{a}_0 + \mathrm{a}_1 t + \mathrm{a}_2 \mathrm{t}^2 + \cdots + \mathrm{a}_5 \mathrm{t}^5 \\ s(\mathrm{t}) = \mathrm{b}_0 + \mathrm{b}_1 t + \mathrm{b}_2 \mathrm{t}^2 + \cdots + \mathrm{b}_4 \mathrm{t}^4 \end{cases} \tag{3.1}$$

Given the initial state, target state, and the planning horizon, the system solves the coefficients using the following boundary conditions:

$$\begin{cases} l(t=0) = l_s \\ \dot{l}(t=0) = v_{ls} \\ \ddot{l}(t=0) = a_{ls} \\ l(t=\mathrm{T}) = l_e \\ \dot{l}(t=\mathrm{T}) = v_{le} \\ \ddot{l}(t=\mathrm{T}) = a_{le} \end{cases}, \quad \begin{matrix} s(t=0) = s_s \\ \dot{s}(t=0) = v_{ss} \\ \ddot{s}(t=0) = a_{ss} \\ \dot{s}(t=\mathrm{T}) = v_{se} \\ \ddot{s}(t=\mathrm{T}) = a_{se} \end{matrix} \tag{3.2}$$

Where l_s, v_{ls}, a_{ls} and l_e, v_{le}, a_{le} represent the lateral position, velocity, and acceleration at the start and end points, respectively. s_s represent the longitudinal position at the start points. Similarly, v_{ss}, a_{ss} and v_{se}, a_{se} represent the longitudinal velocity and acceleration at the start and end points, respectively.

3.1.2 Optimal Trajectory Search based on Cost Function

To generate trajectories that reflect specific driving preferences, this study employs an optimal search method based on a trajectory cost function. Rather than targeting a single fixed state, the algorithm evaluates a candidate set of trajectories using a weighted cost model defined as:

$$J = K_J \cdot \tilde{J}_{int} + K_D \cdot \tilde{l}_e + K_V \cdot \tilde{V}_{error} \tag{3.3}$$

In the formula, $\tilde{J}_{int}$ denotes the integral of the square of the normalized jerk, $\tilde{l}_e$ is the normalized terminal lateral offset, and $\tilde{V}_{error}$ represents the normalized error between the terminal velocity and the target velocity. Adjusting the cost function weights K_J, K_D, K_V can generate trajectories with distinct styles; since $K_V = 1 - K_D - K_J$, practically only multiple $[K_D, K_J]$ combinations need to be set to construct different trajectories through a sampling-optimization approach.

Figure 3.1 and Figure 3.2 illustrate the resulting path and velocity profiles generated under different weight combinations. This approach allows for a flexible representation of driving behavior, where a preferred style is the result of balancing multiple performance metrics rather than adhering to rigid kinematic targets

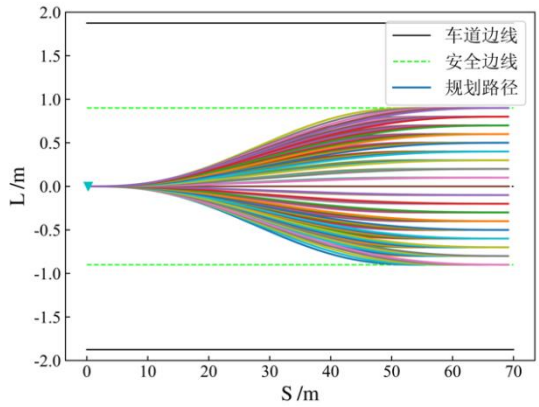


Figure 3.1 Path planning results based on different cost function weights

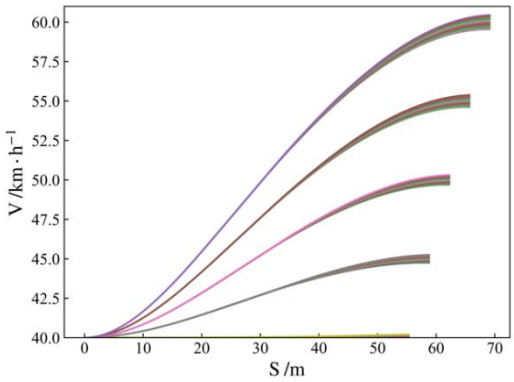


Figure 3.2 Velocity planning results based on different cost function weights

3.2 Trajectory Tracking

The trajectory tracking function adopted in this project is based on the well examined Stanly controller developed by Stanford University in the DARPA Grand Challenge 2005, which realized root mean square cross track error of 0.1m on the real vehicle and under complex circumstances.

3.2.1 Lateral control input and control law

Based on the tracking error information of the vehicle relative to the desired path to be tracked, the controller provides desired steering angle information to be executed and realized by the steering system on the vehicle.

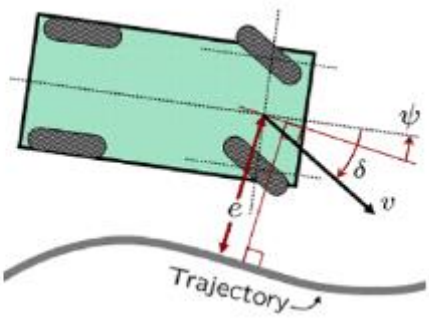


Figure 3.3 Vehicle and Road-Trajectory

As the picture above depicted, the tracking error information is calculated based on a kinematic model, which includes the cross-track error $e(t)$ and heading angle error $\psi(t)$. The cross-track error means the minimal distance from the center point of the front axle (and also the driving and steering axle of the experiment vehicles) to the desired trajectory. The heading angle error means the difference between the heading

angle of vehicle itself and the heading angle of the nearest point on the desired trajectory.

The diagram below shows how these kinds of "preview information" is calculated. Apart from the cross-track error $e(t)$ and heading angle error $\psi(t)$, the curvature of nearest point on the trajectory $\rho_{traj}(t)$ is provided for the feedforward term in the complete control law of lateral controller and desired speed $v_{des}(t)$ is attained for the longitudinal speed controller.

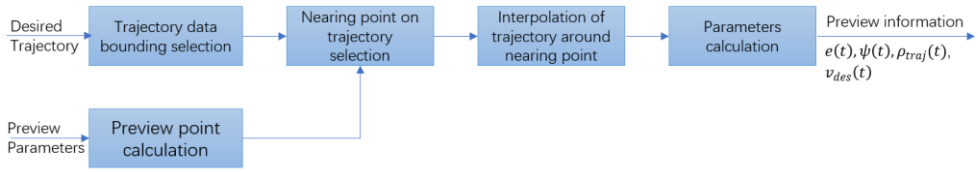


Figure 3.4 Preview information calculation method

The basic control law of the Stanly controller then come as the follow equation, where $v(t)$ means current vehicle velocity. This control law has the advantage that for ideal kinematic models it results in a closed loop system with a globally asymptotically stable equilibrium at $e = 0$ for $v(t) > 0$ and $0 < \delta_{max} < \frac{\pi}{2}$. That is to say, the vehicle motion will always tend to converge to the desired trajectory when controller parameters are well tuned despite the initial state of the vehicle, which provides a promising basis for the motion stability of the vehicle when carry out experiments.

$$\delta_1(t) = \begin{cases} \psi(t) + arctan\dfrac{ke(t)}{v(t)}, & \left|\psi(t) + \arctan\dfrac{ke(t)}{v(t)}\right| < \delta_{max} \\ \delta_{max}, & \psi(t) + arctan\dfrac{ke(t)}{v(t)} \geq \delta_{max} \\ -\delta_{max}, & \psi(t) + arctan\dfrac{ke(t)}{v(t)} \leq -\delta_{max} \end{cases} \tag{3.4}$$

Controller command is provided to the steering system. The complete control law of the controller $\delta_3(t)$ used in this project is shown in the equation below. A feedforward term from the trajectory curvature is added to the original control law to improve the transient performance of the whole system. Time delay and overshoot in the steering system can cause instability. To prevent relative unwanted vehicle motion, an PD controller is combined with the Stanly controller to improve the dynamic performance of the vehicle system, which take similar effects of additional damping item of steering angle and vehicle yaw rate item in the original Stanly controller paper[34].

$$\delta_2(t) = \delta_1(t) + K_{feedforward}\rho_{traj}(t) \tag{3.5}$$

$$\delta_3(t) = K_P\delta_2(t) + K_d\dot{\delta}_2(t) \tag{3.6}$$

3.2.2 Lateral control convergence proof

A brief proof was listed here according to the original Stanly controller paper. Substitute the basic control law for steering angle $\delta_1(t)$ into the kinematic model of an experiment vehicle shown in the equation below, where L is the wheelbase of the vehicle.

$$\begin{cases} \dot{e}(t) = v(t)\sin\left(\psi(t) - \delta(t)\right) \\ \dot{\psi}(t) = r(t) = -\dfrac{v(t)\sin\left(\delta(t)\right)}{L} \end{cases} \tag{3.7}$$

Then we can get the expression of the cross-track error in the future as shown below.

$$\dot{e}(t) = -v(t)\sin\arctan\left(\frac{ke(t)}{v(t)}\right) = \frac{-ke(t)}{\sqrt{1+\left(\dfrac{ke(t)}{v(t)}\right)^2}} \tag{3.8}$$

It can be seen from the formula that when the controller works inside the nominal region, the sign of cross-track error will always be opposite to the sign of its derivative for any $e(t), |\dot{e}(t)| \in [ke(t), v(t)]$. The convergence rate of cross-track error is between linear, with rate $v(t)$ and exponential with rate k. The original paper [35] also proved the convergence of the control law even when the max steering angle δ_{max} is reached and the point $\psi(t) = 0, e(t) = 0$ is the only stable equilibrium in phase space, which guarantees reliable tracking performance when adopting this controller in our experiments.

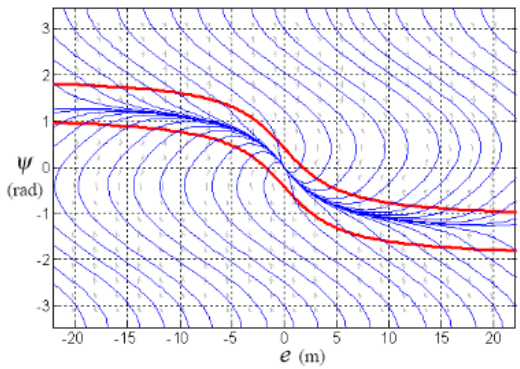


Figure 3.5 The correlation between heading error and cross-track error

3.2.3 Longitudinal control law

For experiments that take place in JTEKT, existed vehicle speed controllers are adopted to control the longitudinal vehicle velocities. For experiments carried out by the Tongji University, the commonly used PD controller is adopted.

Given the desired vehicle speed $v_{des}(t)$ and real vehicle longitudinal speed $v_{veh}(t)$, the positions of brake pedal $p_{brk}(t)$ and acceleration pedal $p_{acc}(t)$ are

generated in the manner shown in the equations below, where K_{pv} and K_{dv} are the parameters of the ACC controller to be tuned.

$$
\begin{aligned}
v_d(t) &= K_{pv}\left(v_{des}(t)-v_{veh}(t)\right)+K_{dv}\left(\dot{v}_{des}(t)-\dot{v}_{veh}(t)\right) \\
p_{brk}(t) &= \begin{cases} 0, & v_d(t)\geq 0 \\ -k_{brk}v_d(t), & v_d(t)<0 \end{cases} \\
p_{acc}(t) &= \begin{cases} k_{acc}v_d(t), & v_d(t)\geq 0 \\ 0 & v_d(t)<0 \end{cases}
\end{aligned}
\tag{3.9}
$$

3.2.4 Validation of the trajectory tracking performance

After the development of trajectory tracking controller, we validated the performance of it by a Hardware-in-the-Loop experiment on the Tongji EPS test-bench. The experiment scenario was the same with final test-bench experiments and the real vehicle experiment carried out at JTEKT and Tongji University. The desired trajectories were generated by a simplified online adaptation system.

Figure 3.6 describes the cross-track error $e(t)$ performance of the trajectory tracking controller. When following all the generated desired trajectories on the curve road, the cross-track error (also usually named, lateral offset error) was controlled below $\pm$5cm.

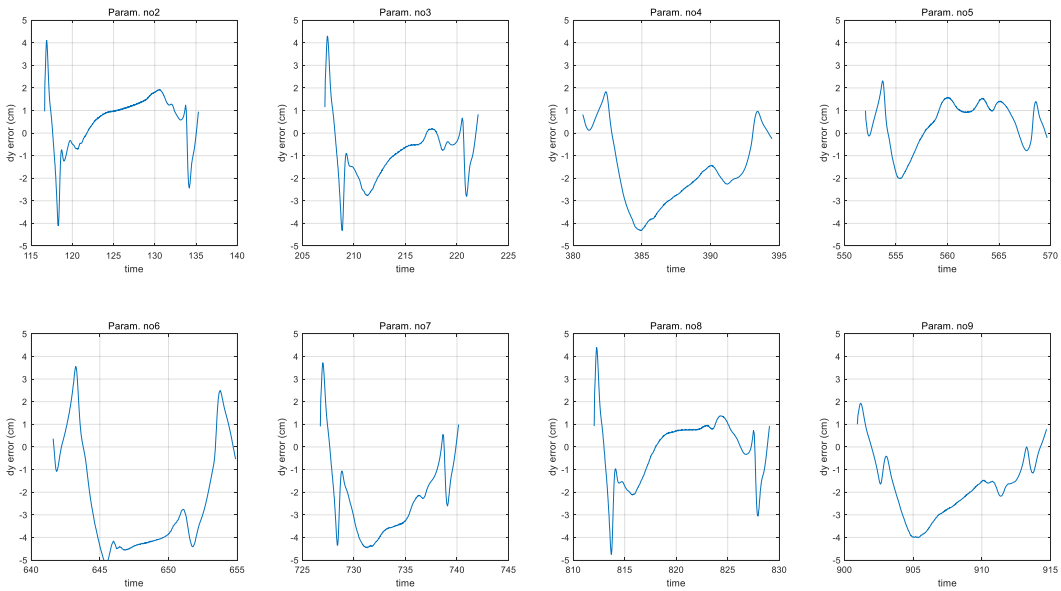


Figure 3.6 The cross-track error performance of the trajectory tracking controller

Figure 3.7 below shows the yaw angle error of the designed tracking controller. The term yaw angle error here means the difference of the yaw angle of preview point of Stanly controller (in the middle of the front axle) and the yaw angle of the nearest point on the trajectory to that point. It is to be noticed that yaw angle error differs from the heading angle error $\psi(t)$ inputted to the controller with the latter

considering heading angle error not yaw angle error. It can be seen from the figure that the yaw angle error was controlled below ±0.8° .

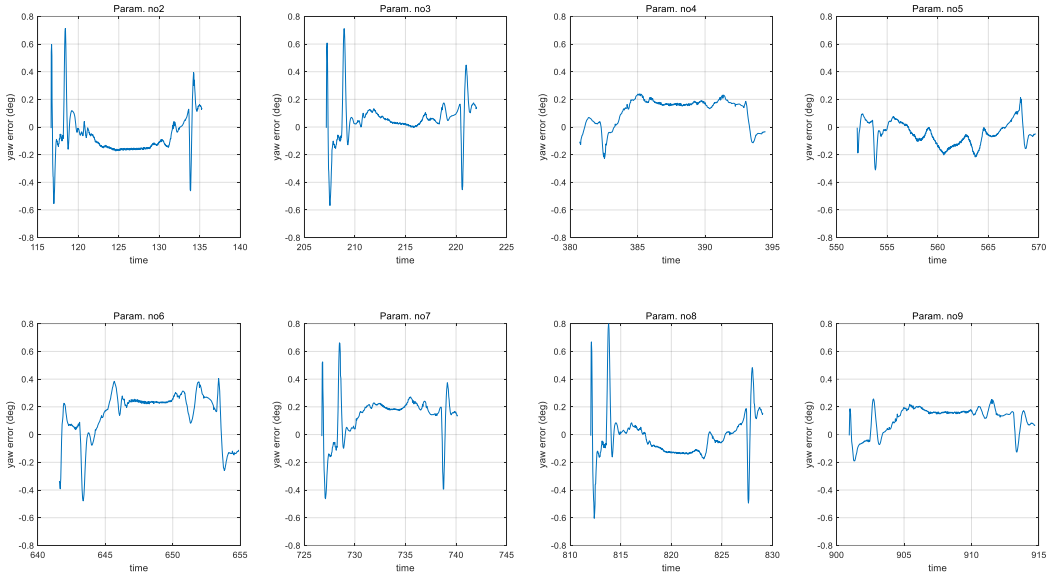


Figure 3.7 The yaw angle error performance of the trajectory tracking controller

Figure 3.8 below presents the longitudinal speed control performance of the trajectory tracking controller, in which the red lines show the desired longitudinal speed of trajectories, and the blue lines show the real speed of the vehicle. As is shown that the longitudinal speed error was controlled below around ±2km/h.

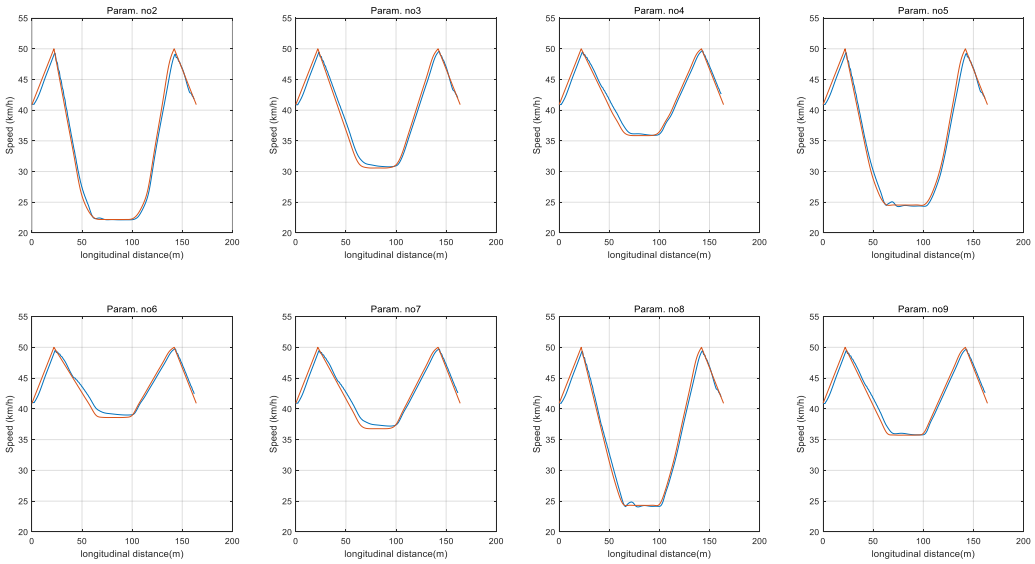


Figure 3.8 The longitudinal speed control performance of the trajectory tracking controller

In conclusion of these results, the designed trajectory following controller meets the needs of experiments.

4 Online adaptive system design

The preceding research established a robust foundation by identifying a set of quantifiable preference indicators, such as longitudinal jerk, lateral acceleration, and developing a high-fidelity trajectory planning algorithm capable of adjusting its behavior via tunable parameters. However, even with these tools in hand, a critical scientific question remains: How can we systematically design a trajectory generation process that does not merely follow a rigid safety protocol, but proactively aligns with the nuanced, subjective preferences of an individual driver?

To address this challenge, this report presents a comprehensive algorithmic framework designed to learn driver preferences directly from trajectory data and adapt the vehicle's behavior in real-time. We develop a three-stage evolutionary path: learning universal reward function through inverse reinforcement learning (IRL), initializing a generalized policy using this reward function, and evolving into a personalized policy via online intervention-based learning. The following sections detail the mathematical and experimental validation of this adaptive system design.

4.1 Reward function design

4.1.1 Coefficients learning framework

We focus on learning reward functions that are consistent with the subjective feelings of drivers and building a parameter decision framework based on the reward function. Inverse reinforcement learning (IRL) methods aim to learn underlying reward functions from expert demonstrations. Many IRL algorithms have been applied to learn driving styles and maneuver preferences, including the maximum margin method and maximum entropy method[36-37]. In order to explicitly represent and interpret human-preferred driving behaviour, we employ the maximum entropy method to adjust the weights of reward function.

The study scenario in this paper is a two-lane curved road with an obstacle vehicle in the neighbouring lane in front of the ego vehicle. The proposed coefficients learning framework is illustrated in Figure 4.1 and consists of three core modules: parametric trajectory generation module, trajectory features extraction module, and trajectory evaluation module. The parametric trajectory generation module synthesizes a diverse set of trajectory proposals by sparsely sampling plausible maneuvers and velocity profiles together with known expert trajectories. The trajectory features extraction module calculates the feature values that affect subjective human perception. In the downstream trajectory evaluation module, the related features of different trajectory proposals over the time horizon are combined with learned weights to calculate the rewards and the probabilities of trajectory proposals according to maximum entropy IRL. The trajectory proposal with the highest probability can be used as a reference trajectory for the autonomous vehicle to follow.

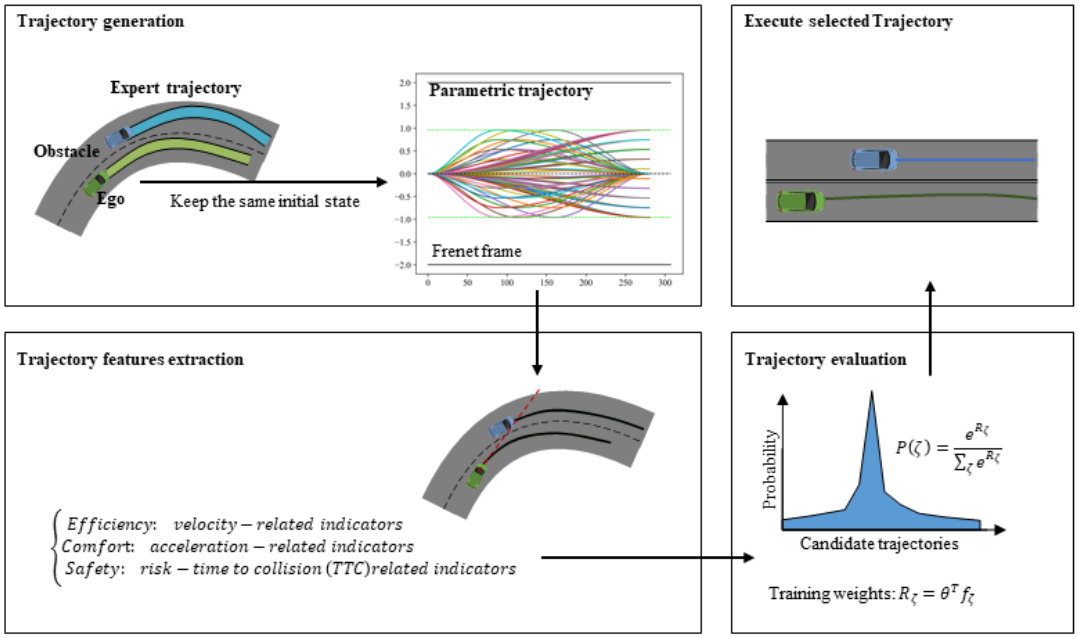


Figure 4.1 The learning framework for reward-function coefficients

4.1.2 Learning from expert trajectories

The learning framework proposed in this research is based on a structural assumption that human decision-making behaviour is focused on high-level intentions instead of low-level control actions, which is consistent with Figure 4.1 Human decision-making behavior under this assumption is divided into three processes, namely trajectory generation, performing pre-judgments of major trajectory features, and trajectory evaluation to select the trajectory best matches human preferences. Given a driving scenario, a human driver first creates multiple candidate trajectory proposals in mind as well as anticipates the features of each trajectory proposal in terms of safety, comfort, and efficiency. Finally, the returns of different trajectory proposals are evaluated using drivers' internal reward functions. The underlying evaluation principle is obtained according to the Boltzmann noisily-rational model[38], whereby the probability of a trajectory is exponential to the reward of the trajectory, and finally, the driver would execute one of the trajectory proposals subject to the distribution. This probabilistic model is plausible and intuitive, as it explains well the stochastic human decision-making behavior and can deal with sub-optimal and multi-optimal strategies existing in naturalistic human decision-making situations.

We assume that drivers' preferences would not change too much over time during a one-time curve drive, so the weights of the reward function are consistent. We suppose a linearized reward function to ensure the results are interpretable. Therefore, the reward function $r(s_t)$ at a specific state s_t is defined as:

$$r(s_t) = \theta^T f(s_t) \tag{4.1}$$

Where $\theta = [\theta_1, \theta_2, ..., \theta_K]$ is the K-dimensional weight vector and $f(s_t) = [f_1(s_t), f_2(s_t), ..., f_K(s_t)]$ is the extracted feature vector that characterizes the state s_t. Therefore, the reward of a trajectory $R(\zeta)$ is given as:

$$R(\zeta) = \sum_t r(s_t) = \theta^T f_\zeta = \theta^T \sum_{s_t \in \zeta} f(s_t) \tag{4.2}$$

Where f_ζ denotes the accumulative features along the trajectory ζ. The selection of features is presented in next section.

Formally, lane-centering is adopted as the expert trajectory to constitute a generalized human-preferred trajectory proposals dataset $\mathcal{D}=\{\zeta_1,\zeta_2,...,\zeta_n\}$ consisting of n trajectories, the problem is to obtain the weights of reward function that can generate a trajectory proposal to match human preferences. We use the maximum entropy IRL algorithm to infer the reward weights θ with the help of our proposed structural assumption on human decision-making behavior.

After generating trajectory proposals and computing the features of these proposals, we can then use the reward function to evaluate drivers' preferences for trajectories. According to the principle of maximum entropy, the probability of human preference for a particular trajectory $P(\zeta_i)$ can be expressed as:

$$P(\zeta_i)=\frac{e^{R(\zeta_i)}}{\sum_j^M e^{R(\tilde{\zeta}_j)}}=\frac{e^{\theta^T f_{\zeta_i}}}{\sum_j^M e^{\theta^T f_{\tilde{\zeta}_j}}} \tag{4.3}$$

Where ζ_i is a trajectory in the dataset $\mathcal{D}$, which contains human-preferred trajectories, $R(\zeta_i)$ is the return of that trajectory. M is the number of trajectory proposals generated by trajectory generation module, $\tilde{\zeta}_j$ is one of the generated trajectories that share the same initial state with ζ_i, and $R(\tilde{\zeta}_j)$ is the return of trajectory $\tilde{\zeta}_j$.

The goal of maximum entropy IRL is to adjust the weights θ to maximize the log-likelihood of human-preferred trajectories $\zeta^*\in\mathcal{D}$:

$$\max_\theta \mathcal{J}(\theta)= \max_\theta \sum_{\zeta\in\mathcal{D}} \log P(\zeta) \tag{4.4}$$

We use gradient-based optimization method with automatic differentiation tools to optimize the weights. To prevent overfitting, we add $L2$ regularization on the weights into the objective function:

$$\mathcal{J}(\theta) = \sum_{\zeta \in \mathcal{D}} \left[\theta^T f_\zeta - \log \sum_{j=1}^{M} e^{\theta^T f_{\tilde{\zeta}_j}} \right] - \lambda \theta^2 \tag{4.5}$$

where $\lambda > 0$ is the regularization parameter.

The gradient becomes the difference of the feature expectations plus the gradient of the regularization term:

$$\nabla_\theta \mathcal{J}(\theta) = \sum_{\zeta \in \mathcal{D}} \left[f_\zeta - \sum_{j=1}^{M} P\left(\tilde{\zeta}_i\right) f_{\tilde{\zeta}_i} \right] - 2\lambda\theta \tag{4.6}$$

4.1.3 Trajectory features extraction

Drivers usually make decisions about preferred trajectories by combining efficiency, comfort, and safety of trajectories. It is assumed in our decision-making frame that human drivers evaluate trajectories according to a certain internal criterion. In order to maximize the consistency between the reward function and drivers' evaluation criteria, we conducted subjective experiments to identify the trajectory features that influence drivers' subjective evaluations[39]. We can compute the features for each candidate trajectory, and the return $R(\zeta)$ of the trajectory is the sum of these features multiplied by the corresponding learnable reward weights. The designed features as described as follows.

1. Travel efficiency: efficiency is defined as the difference between the current speed v and speed limit v_{limit} in time horizon T. This feature is designed to reflect the human's desire to reach the target position as fast as possible.

$$f_{eff}(\zeta) = \frac{1}{T} \sum_{t=1}^{T} |v(t) - v_{limit}| \tag{4.7}$$

2. Comfort: features that represent comfort are the maximum longitudinal acceleration a_x, the maximum lateral acceleration a_y, and the maximum longitudinal jerk $\dot{a}_x$:

$$\begin{cases} f_{com_{ax}}(\zeta) = \max_t |a_x(t)| \\ f_{com_{ay}}(\zeta) = \max_t |a_y(t)| \\ f_{com_{jx}}(\zeta) = \max_t |\dot{a}_x(t)| \end{cases} \tag{4.8}$$

3. Safety: human-preferred trajectories keep a certain distance from surrounding vehicles and lane edges. We define safety as the negative value of risk in reference[40] and add a feature that considers the risk of neighboring vehicles based on the subjective evaluation experiment which comes from another of our research programs.

$$f_{safety}(\zeta) = -risk \tag{4.9}$$

All the above features are calculated at every timestep and accumulated over horizon T to obtain the feature of a trajectory. The features are then normalized to $[0, 1]$ by dividing the maximum value in the dataset to remove the impact of their different units and scales.

4.1.4 Results

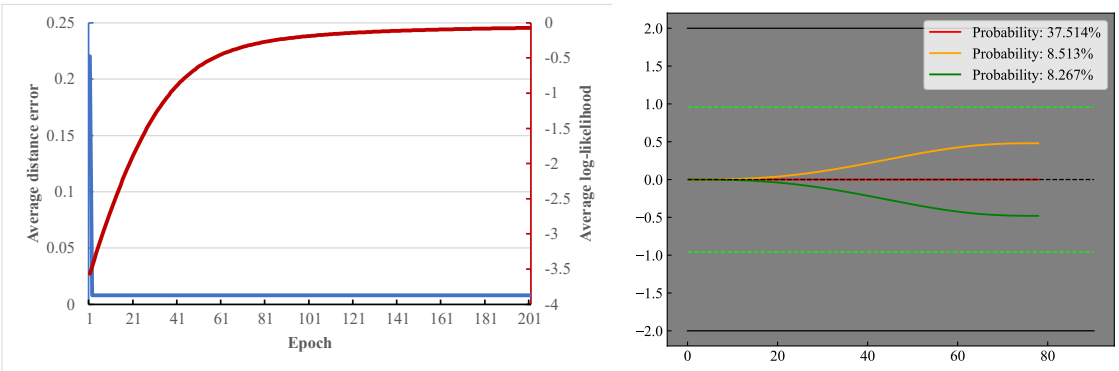
We use the proposed coefficients learning framework to analyze drivers' preferences for curved trajectories. To evaluate the preference prediction performance, two established metrics for prediction results are used. The average distance error (ADE), refers to the average distance between the trajectories given by the algorithm and the ground truth. The prediction accuracy (PA), indicates the probability that top-3 most likely planned trajectories contain the ground truth.

We consider a fixed time horizon $T = 5s$ for trajectory generation module because it can well balance prediction accuracy and long-term decisions and satisfy the requirement of trajectory planning algorithms. We design a simulated driver and

then randomly select start states throughout the curve road. The simulated driver can give the preferred trajectory under a position of the curve, and these trajectories with 5s length of time eventually form a dataset of human-preferred trajectories on curve road. For simplicity, the simulated driver's preferred trajectory in any state is set to be the centerline of the lane with a speed equal to the road speed limit. The result is:

$$\theta = [-6.12, -5.98, -5.32, -5.89, 1.41] \tag{4.10}$$

The training results of the framework applied to the trajectory planning method and overtaking point planning method are shown in Figure 4.2. a) show the reward function learning process, which plots the curves of the average distance error (ADE) between the learned trajectory and human preferred trajectory, and the average log-likelihood of the preferred trajectories. The optimal trajectory selected by the trajectory evaluation module is almost identical to the simulated driver's preferred trajectory. b) show the results of the trajectory evaluation. The candidate trajectory with the top 3 probabilities of preference and the corresponding probability values as well as the ground-truth trajectory are shown. The red trajectory represents the optimal trajectory selected by the algorithm, which has a much higher probability of being selected than the other trajectories.



a) The reward function learning process

b) Evaluation result

Figure 4.2 Training result of reward function learning

The learned reward function is then used to determine the probability of candidate trajectory proposals in a test set. We construct the test set according to 30 different initial states to analyze the accuracy of our proposed framework. The results show that the prediction accuracy (PA) of our framework achieves 97.8%.

4.2 Online Personalized framework

4.2.1 Framework introduction

Learning-based online adaptive driving policies hold considerable promise for enabling human-preferred autonomous driving or advanced driver assistance systems. However, existing personalization approaches often rely on limited style definitions or behavior cloning, requiring extensive data and placing cognitive burdens on users. To address these limitations, an Online Personalized Policy Adaptation (OPPA) framework is proposed based on direct driver interventions, where real-time corrective actions on the steering wheel and pedals are treated as implicit feedback reflecting individual driving preferences. A hybrid learning strategy integrates reinforcement learning with intervention-guided updates, enabling efficient online adaptation. The framework is initialized from a generalized policy pretrained using the twin delayed deep deterministic policy gradient (TD3) algorithm and a composite reward function that captures perceived risk, comfort, and efficiency. Driver-in-the-loop experiments incorporating subjective evaluations are conducted with 21 drivers across four representative curved-lane scenarios to assess the effectiveness of the personalization.

Figure 4.3 illustrates the overall structure of the proposed OPPA framework, including offline general policy learning, online policy updates with human intervention, and interaction with the environment through trajectory planning and tracking. The framework adapts driving policies based on real-time corrective actions, without requiring explicit labeling or manual configuration. A generalized policy is pretrained offline via TD3 with a reward function capturing perceived risk, comfort, and efficiency. During online execution, real-time driver interventions such as adjustments to the steering wheel or pedals are treated as implicit preference feedback.

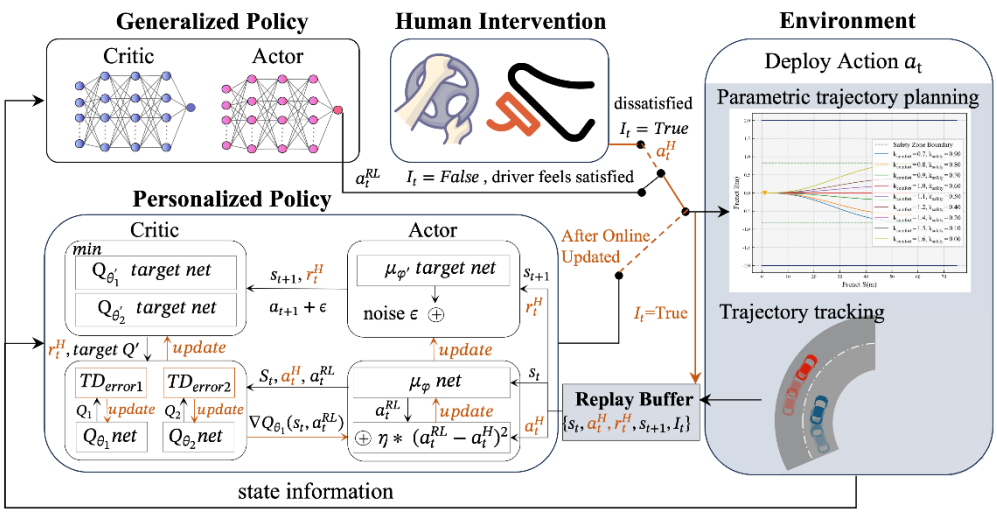


Figure 4.3 Overview of the proposed OPPA framework.

4.2.2 Human-machine co-driving module

The proposed OPPA framework integrates offline pretraining and online adaptation to enable personalized driving policy learning. As shown in Figure 4.4, the generalized policy module outputs trajectory planning parameters, which are then used to generate a reference speed v_{ref} and path ζ_{ref}. The vehicle follows this reference unless driver intervention occurs. When driver intervention is detected ($I_t = 1$), the human-machine co-driving (HMCD) module blends driver's input δ_{sw}^H and pedal depth d^H with the policy outputs δ_{sw}^{RL} based on a coefficient G_{RL}, yielding the final vehicle control commands: speed v and front-wheel steering angle δ_f. Environmental observations s_t, s_{t+1} and r_t are provided for policy adaptation. The transition tuples $\{s_t, a_t, r_t, s_{t+1}, I_t\}$ are then stored in the replay buffer for learning.

When driver intervention is detected ($I_t = 1$), the HMCD module fuses human and policy commands at the control layer:

$$u_t = G_{RL} u_t^{RL} + (1 - G_{RL}) u_t^H \tag{4.11}$$

where u_t is the steering or speed command, $G_{RL} \in [0,1]$ denotes machine authority and $(1 - G_{RL})$ denotes human authority. When $I_t = 0$, the vehicle follows the policy reference by setting $G_{RL} = 1$. Under intervention $I_t = 1$, G_{RL} decreases toward 0.

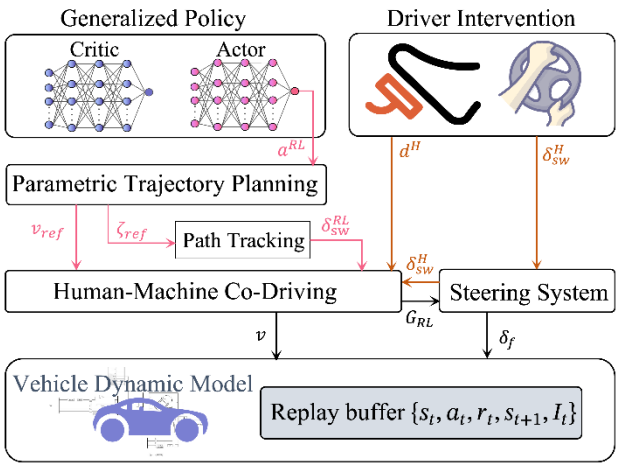


Figure 4.4 The architecture of the human-machine co-driving system.

4.2.3 Problem Formulation

This study formulates the decision process for trajectory planning parameters as a Markov Decision Process (MDP) solved using reinforcement learning (RL) At each time step t, the agent observes state $s_t \in S$, execute action $a_t \in A$, receives reward r_t, and transitions to s_{t+1}. The objective is to maximize the expected discounted cumulative reward $\sum_{t=0}^{\infty} \gamma^t \mathrm{r_t}$, where $\gamma \in (0,1]$ is the discounted factor. Specifically, the state space, action space, and reward function are defined as follows.

1) State Space: At each time step, the state $s_t \in S$ consists of the position and velocity information of the ego vehicle and the state information of adjacent vehicles. It defined as:

$$s_t = [{s_t}^{adj}, {s_t}^{ego}] \tag{4.12}$$

where $s_t^{adj} = [x_i^{adj}, y_i^{adj}, yaw_i^{adj}, {v_i}^{adj}]$, i ranges from 1 to N, where N represents the number of adjacent vehicles. Ego vehicle's information additionally includes longitudinal and lateral accelerations in the Frenet coordinate system[41], thus

$$s_t^{ego} = [x^{ego}, y^{ego}, yaw^{ego}, v^{ego}, lon_{\mathrm{acc}}^{ego}, lat_{\mathrm{acc}}^{ego}].$$

2) Action Space: The action space $a_t \in A$ represents the trajectory planning weights for comfort and safety:

$$a_{\mathrm{t}} = [k_{\mathrm{comfort}}, k_{\mathrm{safety}}] \tag{4.13}$$

with $k_{comfort}, k_{safety} \in [0,1]$, and the efficiency weight implicitly defined to ensure the weights sum to one. These parameters govern planning behavior, with higher comfort weights resulting in smoother but more deviated trajectories, whereas higher safety weights strengthen lane-centering.

3) Reward Function: The reward function evaluates driver preferences over candidate trajectories and is designed as a

weighted sum of five trajectory-related metrics:

$$r_t = w_1 c_{ef} + w_2 c_{lon_{\mathrm{a}}} + w_3 c_{lon_{jk}} + w_4 c_{lat_a} + w_5 c_{\mathrm{rp}} \tag{4.14}$$

where the weights $w = [-6.12, -5.98, -5.32, -5.89, 1.41]$ are obtained via inverse reinforcement learning (IRL) based on expert trajectories. Each trajectory is denoted as ζ, where $\zeta = \{x_j, y_j, yaw_j, v_j\}$ and j denotes the index of trajectory point. Maximizing the reward function requires minimizing efficiency term c_{ef} and comfort terms $c_{lon_{\mathrm{a}}}, c_{lon_{jk}}, c_{lat_a}$ while allowing higher values for risk perception term c_{rp}. Specifically, $c_{ef} = max(|v_j - v_{set}|)$ penalizes deviation from the target speed v_{set}. The comfort-related cost c_{lon_a}, c_{lon_j} and c_{lat_a} denote the maximum absolute longitudinal acceleration, jerk, and lateral acceleration, respectively, along the trajectory ζ. The risk perception cost c_{rp} depends on the ego vehicle's lane position and proximity to adjacent vehicles and boundaries. When in the inner lane:

$$c_{rp} = ictad + 0.16 \cdot (2.17 \cdot istlc + 0.33 \cdot ittc) \tag{4.15}$$

when ego vehicle is traveling in the outer lane, it is given by:

$$c_{rp} = ictlc + 0.08 \cdot (1.09 \cdot istlc + 0.45 \cdot ittc) \tag{4.16}$$

where the intermediate metric defined as:

$ictad = v_{c-relative}/d_{anticipate}$, where $v_{c-relative}$ is the relative velocity tangential to the adjacent vehicle's predicted path, and $d_{acticipate}$ is the curved distance to the potential collision point assuming constant speed and steering.

$ictlc = v_{ego}/d_{curved}$, where v_{ego} is the speed of the ego vehicle and d_{curved} denotes the curved distance needed to cross the lane boundary given current steering.

$istlc = v_{ego}/d_{straight}$, where $d_{straight}$ is the distance to cross the lane boundary at zero steering.

$ittc = v_{relative}/d_{relative}$, where $d_{relative}$ is the relative distance between the ego and adjacent vehicles, and $v_{relative}$ is its time derivative, representing the closing speed.

4.3 Offline Driver preference model building

To provide a safe and generalizable initialization for subsequent online personalization, a base policy is first trained offline using the TD3 algorithm combined with Prioritized Experience Replay (PER). This pretraining stage ensures that the system exhibits reasonable driving behavior before encountering any driver-specific preferences, thereby accelerating convergence during online adaptation and reducing the reliance on early human interventions.

In TD3, the agent selects actions through a deterministic actor network $\mu(s)$, while two critic networks $Q_1(s,a)$ and $Q_2(s,a)$ estimate the action value. To enhance training stability, TD3 maintains a set of target networks: μ' for the actor and Q_1', Q_2' for the critics, which are updated using soft target updates to ensure smooth convergence. The target Q-value is computed as:

$$y_t = r + \gamma min_{i=1,2} Q_{\theta_i'}\left(s_{t+1}, \mu_{\phi'}'(s_{t+1}) + \epsilon\right) \tag{4.17}$$

where θ_i' and ϕ' represent the parameters of the target critic and actor networks. The term $\epsilon \sim \mathrm{clip}(\mathcal{N}(0,\sigma), -c, c)$ is clipped Gaussian noise used to smooth the target policy, constrained within the interval $[-c, c]$.

The temporal-difference (TD) error is defined as the difference between the predicted Q-value and the target Q-value. The critic networks minimize the mean squared TD error:

$$L^{Q_i}(\theta_{i=1,2}) = E_{(s_t,a_t,r_t,s_{t+1})\sim\mathcal{D}}\left[\left(Q_{\theta_i}(s_t,a_t) - y_t\right)^2\right] \tag{4.18}$$

where E denotes the expectation over the mini-batch sampled from the replay buffer $\mathcal{D}$, and θ_i represents the parameters of the critic network.

The objective of the actor network is to maximize the estimated Q-value of the action:

$$L^{\mu}{}_{TD3}(\phi) = -E_{s_t \sim \mathcal{D}}\left[Q_{\theta_1}\left(s_t, \mu_\phi(s_t)\right)\right] \tag{4.19}$$

where ϕ denotes the parameters of the actor network.

To accelerate learning, PER is employed, which prioritizes transitions based on their TD errors. The sampling probability for each transition j is calculated as:

$$P_j = p_j^{\rho} / \sum p^{\rho} \tag{4.20}$$

where p_j is proportional to the TD error of transition j, and $\rho \in [0,1]$ controls the degree of prioritization. To correct the bias introduced by non-uniform sampling, importance-sampling (IS) weights are applied:

$$\omega_j = \left(\frac{P_j}{\min_k P_k}\right)^{-\beta} \tag{4.21}$$

where $\beta \in [0,1]$ adjusts the strength of correction. The final PER-weighted critic loss becomes:

$$L^{Q_i}(\theta_{i=1,2}) = \mathrm{E}\left[\omega\left(Q_{\theta_i}(s_t, a_t) - y_t\right)^2\right] \tag{4.22}$$

The resulting actor network defines the generalized policy μ_{RL}, which captures generalizable driving behaviors and serves as the basis for online adaptation.

During offline training, PER yielded consistently lower actor loss and higher average episodic reward than vanilla TD3 (120.44 ± 39.45 vs. 50.87 ± 67.88, over 5 seeds; each episode corresponds to a complete curve traversal), indicating more stable, sample-efficient learning and providing a stronger initialization for subsequent online personalization.

4.4 Online Personalized Policy Adaptation

Personalized driving preferences vary across drivers and scenarios, potentially causing mismatches between the generalized policy μ_{RL} and individual expectations. To address this, a personalized policy μ_P is adapted from μ_{RL} using the driver $\mathcal{B}^H$:

$$\mu_P = f_p(\mu_{RL}, \mathcal{B}^{\mathrm{H}}) \tag{4.23}$$

where $f_p(\cdot)$ represents the adaptation process using driver intervention data to refine the base policy.

The goal is to improve preference alignment, ensuring that $Pref[\mu_P] \geq Pref[\mu_{RL}]$. This is later verified through driver-in-the-loop evaluations using subjective preference scores.

During execution, the policy first outputs an action a_t^{RL} from μ_{RL}. If the driver disagrees with the behavior, they intervene by operating the steering or pedals. The system records the intervention flag $I_t \in \{0,1\}$, and infers a corrective action a_t^H. The blended control action is:

$$a_t = I_t a_t^H + (1 - I_t) a_t^{RL} \tag{4.24}$$

Intervention transition $\{s_t, a_t^H, r_t^H, s_{t+1}, I_t\}$ are stored, where $r_t^H = \kappa \cdot r_t$ $(\kappa > 1)$ emphasizes the driver's intended behavior. The critic target is:

$$\hat{y}_t = r_t{}^{\mathrm{H}} + \gamma \min_{i=1,2} Q_{\theta_i'}\left(s_{t+1}, \mu_{\phi'}(s_{t+1}) + \epsilon\right) \tag{4.25}$$

The critic loss under driver intervention is defined as:

$$L^{Q_i}\left(\theta_{i=1,2}\right) = E\left[\left(Q_{\theta_{i=1,2}}(s_t, a_t^{\mathrm{H}}) - \hat{y}_t\right)^2\right] \tag{4.26}$$

To accelerate personalization, a least-squares imitation term is added to the original actor loss:

$$L^{\mu}{}_{\mathrm{TD3}'}(\phi) = L^{\mu}{}_{\mathrm{TD3}}(\phi) + \eta * [\mu_\phi(s_t) - a_t{}^{\mathrm{H}}]^2 \tag{4.27}$$

where η is a dynamic confidence weight that scales the influence of driver interventions. It is defined as:

$$\eta = \gamma^{\mathrm{k}}[max(exp(\alpha \Delta Q), 1) - 1] \tag{4.28}$$

with k denoting the update count and ΔQ is the Q-value advantage of the intervention over policy:

$$\Delta Q = \hat{y}_t - Q_1\left(s_t, \mu_\phi(s_t)\right) \tag{4.29}$$

This formulation emphasizes only those interventions that yield higher expected returns. As shown in Fig. エラー! 参照元が見つかりません。, η decays with k, indicating reduced reliance on interventions over time, and grows with ΔQ, especially under larger confidence factor α.

4.5 Experimental Results

4.5.1 Experiment Setup

The proposed OPPA framework is evaluated through a series of driver-in-the-loop experiments on a fixed-base driving simulator, as shown in Figure 4.5. The simulator consists of four key components: a real-time target machine, a steering system and pedals, a personal computer (PC), and a curved display screen (3840x1080). The PC runs the CARLA simulator and the OPPA framework, including the policy decision and trajectory planning modules. The real-time target machine is responsible for low-level trajectory tracking and vehicle dynamic simulation. It outputs alignment torque commands to the loading motor and provides haptic feedback torque to the Electric Power Steering (EPS) motor. The PC and the real-time target machine communicate via the Controller Area Network (CAN) bus to ensure reliable bidirectional transmission. Specifically, the PC sends control signals to the real-time machine, while the real-time machine transmits simulated vehicle states back to the PC. This closed-loop communication enables seamless integration with the CARLA environment and reinforcement learning-based control logic.

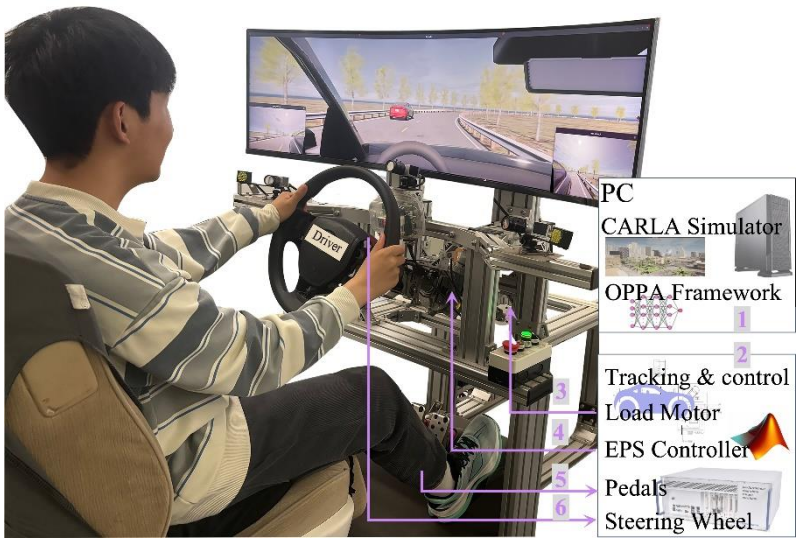


Figure 4.5 Fixed-base driving simulator setup for driver-in-the-loop evaluation.

1) Driving Scenario Descriptions

Drivers exhibit distinct personalized preferences when navigating curves[39]. Therefore, this study focuses on a representative highway ramp involving a right-turn double-lane curve. The lane-restricted ego vehicle travels at 40-60 km/h while an adjacent vehicle maintains a constant speed.

To evaluate the OPPA algorithm's online adaptation under conditions consistent with the reward function, four driving scenarios are designed, as illustrated in Figure 4.6. The scenarios vary by the ego lane (inner or outer) and the adjacent vehicle's relative position, and ego vehicle remains in its lane. Driver interventions serve as preference feedback for adaptation.

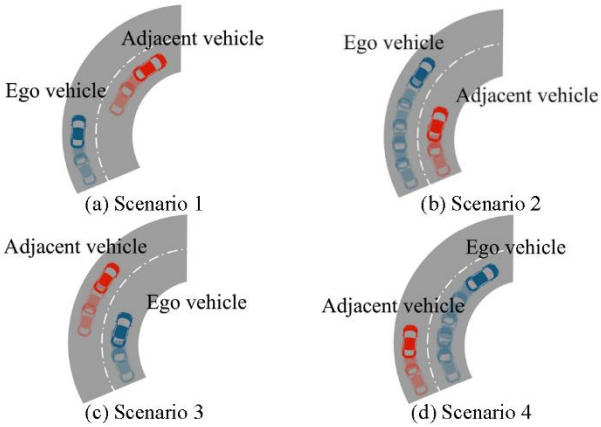


Figure 4.6 Curved-lane driving scenarios used in the driver-in-the-loop experiments.

2) Online Testing with Driver Evaluation

Once the driver has familiarized themselves with the experimental procedure and completed a practice session on the simulator, the formal experiment begins. Initially, the ego vehicle follows the generalized policy μ_{RL}, driving along the lane centerline at 40 km/h. The driver experiences system behavior under generalized μ_{RL} and personalized μ_P across the four curved-lane scenarios, and then provides subjective evaluations using the questionnaire shown in Table 4.1. The questionnaire includes an overall satisfaction score (1-5 Likert scale) and three categorical items assessing safety, comfort, and efficiency (1: *Needs Improvement*, 2: *Acceptable*, 3: *Satisfied*). This numerical scale is used for clarity in the results.

Table 4.1 Subjective evaluation of driving performance and intervention intentions

Evaluation Dimension	**Subjective Evaluation**		
Subjective evaluation for general policy μ_{RL} before intervention			
Safety	Needs Improvement	Acceptable	Satisfied
Efficiency	Needs Improvement	Acceptable	Satisfied
Comfort	Needs Improvement	Acceptable	Satisfied
Overall Preference score		(1-5 scale)	
Subjective evaluation for personalized policy μ_P after intervention			
Safety	Needs Improvement	Acceptable	Satisfied
Efficiency	Needs Improvement	Acceptable	Satisfied
Comfort	Needs Improvement	Acceptable	Satisfied
Overall Preference score		(1-5 scale)	

If the driver is dissatisfied with the trajectory behavior, they intervene using the steering wheel or pedals. These real-time interventions provide implicit feedback, guiding the OPPA algorithm for online policy adaptation. The updated personalized policy μ_p is then deployed in the same scenario, and the driver re-evaluates the system using the same questionnaire. All interventions, vehicle states, and subjective evaluations are recorded for further analysis, comparing preference scores before and after adaptation.

4.5.2 Driver-in-the-loop Intervention Result

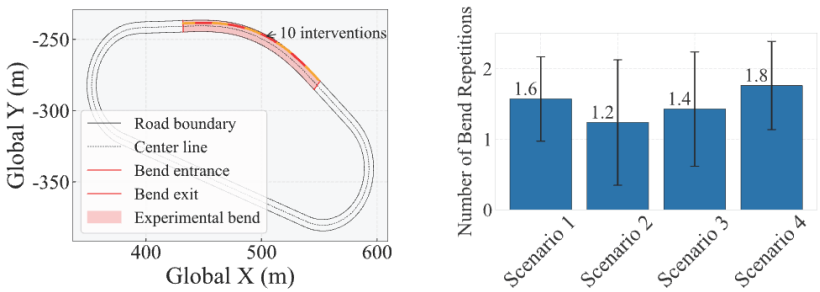
A total of 21 participants, consisting of 11 regular drivers (average driving experience: 3.6 years) and 10 professional drivers (average driving experience: over 19 years), took part in the driver-in-the-loop experiment, as shown in Table 4.2. Each participant experienced four distinct driving scenarios to assess subjective preferences and evaluate policy alignment across different vehicle interaction contexts.

Table 4.2 Driver participant information

No.	**Gender**	**Driving Age**	**Age**	**Driving Frequency**
1	Male	6	25	Regular
2	Male	1	26	Occasional
3	Male	1	27	Occasional
4	Female	5	25	Regular
5	Female	2	26	Occasional
6	Male	3	26	Occasional
7	Male	8	28	Regular
8	Female	6	28	Regular
9	Female	2	28	Occasional

10	Male	4	24	Regular
11	Male	2	23	Occasional
12	Male	21	49	Professional
13	Male	15	55	Professional
14	Male	10	46	Professional
15	Male	20	53	Professional
16	Male	23	50	Professional
17	Male	20	57	Professional
18	Male	19	50	Professional
19	Male	12	55	Professional
20	Male	30	42	Professional
21	Male	20	53	Professional

In the OPPA framework, trajectory points are sampled at 0.02-second intervals, and a driver intervention is defined as the continuous detection of 50 valid intervention signals. As shown in Figure 4.7 (a), the policy begins updating once more than 10 intervention segments are collected. After the generalized policy is updated, the ego vehicle drives under the personalized policy, and the driver is asked to re-evaluate the trajectory. If the personalized trajectory aligns with the driver's preferences and no further improvement is requested, the trial concludes. Figure 4.7 (b) presents the number of repeated bend segments required for the driver to be satisfied with the personalized policy. Results show that, across four scenarios, the average number of required bend repetitions is fewer than two. Across all 21 participants, the average personalization time per intervention is 0.75 ± 0.15 seconds, demonstrating the high adaptation speed of the OPPA framework.



(a) (b)

Figure 4.7 Evaluation of personalization efficiency in driver-in-the-loop experiments.

To evaluate the effect of online personalization, エラー! 参照元が見つかりません。 presents the intervention and adaptation results from a representative driver

across the four designed scenarios. Each subplot illustrates the generalized trajectory before intervention (blue), the driver's intervention trajectory (green), and the personalized trajectory after OPPA adaptation (red). In エラー! 参照元が見つかりません。 (a) scenario 1, the ego vehicle is in the outer lane and slower than the adjacent vehicle; the green trajectory shows that the driver's acceleration intervention and a lateral deviation toward the inner side. The red trajectory demonstrates that the OPPA framework effectively captured this preference and adjusted the policy accordingly. Similar alignments are observed in the remaining scenarios. Notably, in エラー! 参照元が見つかりません。 (d) scenario 4, the driver only adjusts the steering without speed intervention. This highlights a key advantage of intervention-guided learning over imitation learning, as the system requires corrections only when the driver expresses dissatisfaction, thus reducing the cognitive load on the driver.

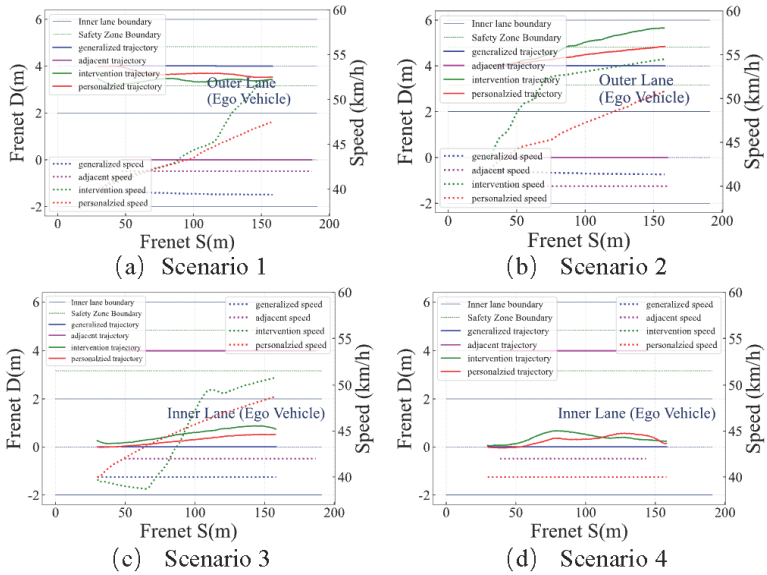


Figure 4.8 Comparison of generalized, intervention, and personalized trajectories from a representative driver across four driving scenarios. Blue: initial generalized policy trajectory; Green: driver intervention trajectory; Red: OPPA-updated personalized trajectory.

Additionally, we quantify the effect across all drivers using objective discrepancies relative to the intervention trajectory, as shown in Figure 4.9, RMS lateral offset difference ($RMS\ Lat.$) and maximum lateral offset differences ($Max\ Lat.$) for safety, RMS speed difference ($RMS\ Speed$) for efficiency, and RMS

longitudinal-jerk difference ($RMS\ Jerk$) for comfort. Across drivers, paired-sample t-tests show that the personalized–intervention discrepancies are significantly smaller than the generalized–intervention discrepancies, indicating alignment with driver preferences and improved tracking accuracy, speed consistency, and motion smoothness.

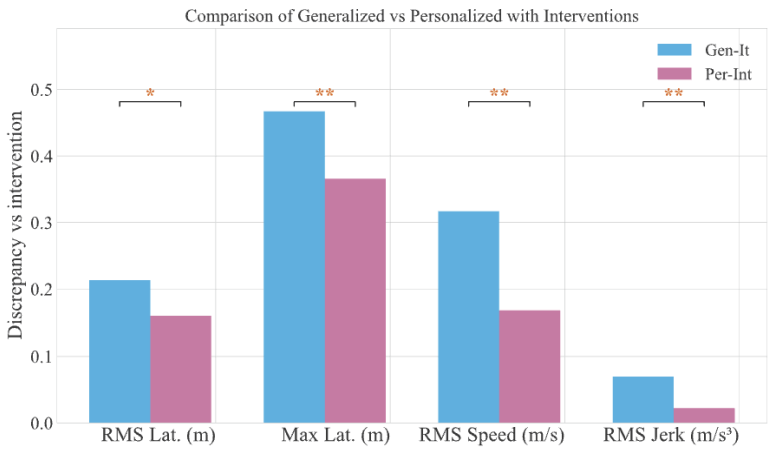


Figure 4.9 Discrepancies relative to the intervention trajectory across all drivers. Gen–Int: generalized vs intervention; Per–Int: personalized vs intervention. Significance levels: *p<0.05, **p<0.01; ns, not significant.

To assess the effectiveness of the OPPA framework, subjective evaluations from 21 participants are analyzed. Paired-sample t-tests compare the scores given before and after policy adaptation across scenarios. In Figure 4.10 (a), subjective ratings for safety, efficiency, and comfort are converted into numerical scores. The results show significant improvements in safety and efficiency, while comfort remains statistically unchanged. This is likely due to the generalized policy already receiving high initial comfort ratings. Figure 4.10 (b) presents satisfaction scores in all four scenarios, where personalized policies consistently receive higher ratings. Figure 4.10 (c) shows the overall score distribution, with personalized policies yielding higher medians and tighter upper ranges. Figure 4.10 (d) illustrates the improvement percentage in each scenario, with the average improvement reaching 18.9%, indicating the alignment of personalized policies with driver preferences. The OPPA algorithm's hyperparameter settings are detailed in Table 4.3.

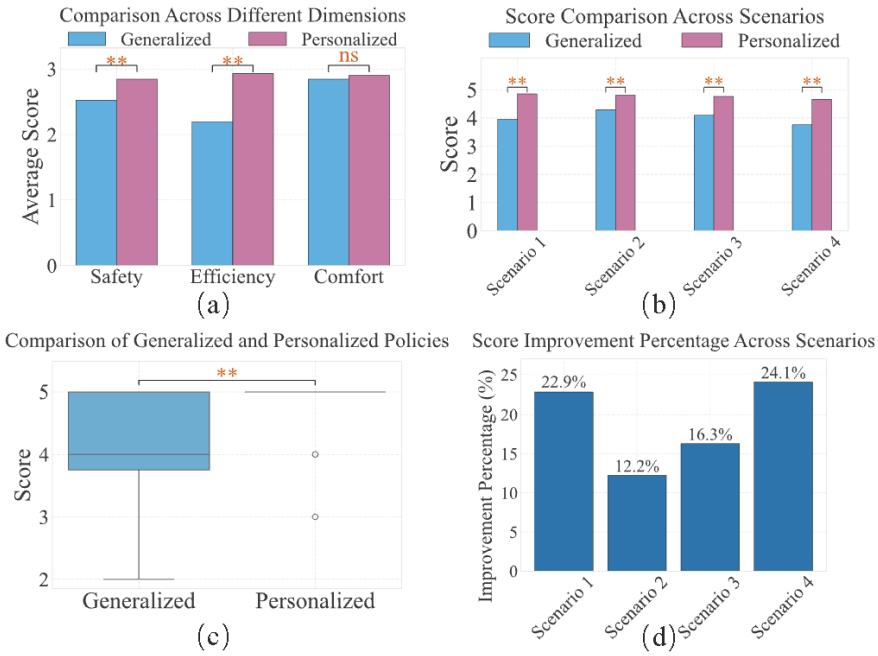


Figure 4.10 Comparison between generalized and personalized policies: (a) Overall score distribution, (b) Performance across safety, efficiency, and comfort dimensions. (c) Score comparison before and after personalization for each driver. (d) Percentage improvement across different scenarios. The results demonstrate significant improvements in the personalized policy, particularly in efficiency and non-congested scenarios. Significance levels: $^{**}p < 0.01$; ns, not significant.

Table 4.3 Hyperparameters setting for the OPPA framework

Parameter	***Explanation***	***Value***
Maximum episode	Cutoff episode number of training	1e3
Replay buffer size	Capacity of ecperience replay buffer	5e3
Minibatch size	Capacity of offline minibatch	100
Minibatch size	Capacity of online minibatch	10
Activation function	Activation funciton of networks	ReLU
Soft update factor	Interpolation to target networks	1e-2
Policy delay	Update frequency of critic over actor	2
Discount factor γ	Discount factor of Bellman equation	0.99
Prioritization ρ	Prioritization degree of PER	0.6
Strength β	Correction strength of PER	0.4
Intervention κ	Multiplies reward during intervention.	5.0
Confidence factor α	Advantage degree of intervention	1.3
Weight λ	Regularization weight of MAS	3.0

4.6 In Conclusion

We propose an Online Personalized Policy Adaptation (OPPA) framework that enables efficient online policy updates using driver interventions as implicit preference feedback. The system architecture is developed through two distinct phases:

1. Offline Policy Initialization, a generalized driving policy is pretrained offline using the TD3 algorithm and a composite reward function that accounts for perceived risk, comfort, and efficiency, ensuring safe and reasonable behavior prior to personalization.

2. Adaptive Online Learning, an online personalized policy adaptation (OPPA) framework is proposed, where real-time driver interventions serve as implicit preference feedback. This phase employs a hybrid learning strategy that integrates standard reinforcement learning with intervention-guided updates.

The framework's efficacy was validated through driver-in-the-loop experiments across four complex curved-lane scenarios. Subjective preference alignment was rigorously evaluated via structured assessments before and after adaptation. Results demonstrate that the personalized policy significantly enhances user experience, improving subjective satisfaction by 18.9% over the baseline. Notably, the system achieves rapid adaptation, with an average per-intervention learning time of only 0.75 ± 0.15 seconds, confirming its suitability for real-time deployment.

5 5. Summary and Future Work

5.1 Summary

This project addressed the challenge of personalizing Lane Centering Control (LCC) systems in complex, dynamic driving environments. Unlike traditional methods that rely on static parameters or simple imitation, we focused on dynamic curved scenarios involving adjacent vehicles, proposing a comprehensive solution

spanning from risk perception quantification to online reinforcement learning control. The key achievements and conclusions are summarized as follows:

1. Quantification of Dynamic Risk Perception: We established a multi-dimensional risk perception metric system distinguishing between Latent Risk (using STLCi and TTCi) and Overt Risk (using CTADi and CTLCi). Through subjective evaluation experiments based on Stevens' Power Law and validation via natural driving data, we derived a quantitative model that accurately maps objective vehicle states to the driver’s subjective psychological safety boundaries. This provided a computable "safety layer" for the control system.
2. Online Adaptive System (OAS) Implementation: We developed an Online Adaptive System based on the TD3 (Twin Delayed Deep Deterministic Policy Gradient) reinforcement learning algorithm. A novel loss function incorporating a "Human Intervention Term" was designed, enabling the agent to learn directly from the driver’s takeover behaviors. This mechanism allows the system to real-time optimize trajectory planning parameters to align with individual driving styles.
3. Experimental Validation and Efficiency: The system was rigorously tested across simulation benches and real-vehicle platforms. Results demonstrate that the OAS can capture driver preferences with high efficiency, requiring only 10 seconds of cumulative intervention time to converge, with a computational update time of 0.75 seconds. Compared to generalized baseline strategies, the personalized system achieved an average 18.9% improvement in comprehensive performance scores (safety, efficiency, and satisfaction), successfully realizing a human-centric driving experience in dynamic curves.

5.2 Future work

With the successful verification of the core algorithms and the prototype system, the research phase of this joint project has reached a solid conclusion. To further advance this technology towards mass production and broader application, the following directions are recommended for future development:

1. Algorithm Embedded Transplantation and Optimization: Currently, the algorithm runs on high-performance industrial computers. Future work should focus on optimizing the TD3 network structure (e.g., model pruning or quantization) to deploy the OAS onto automotive-grade embedded chips (ECUs) with limited computing power. Ensuring real-time performance on production hardware is a critical step for commercialization.
2. Scenario Generalization: While the current system performs excellently in dynamic curves with adjacent vehicles, real-world driving involves more complex variables (e.g., multi-vehicle interaction, intersections, or variable lane widths). Expanding the training dataset and risk metrics to cover a wider range of Operational Design Domains (ODD) will enhance the system's robustness and universality.
3. Long-term Preference Management: The current system excels at "short-term adaptation" for a single trip. Future development could introduce a user profile management module to store learned parameters locally or in the cloud. This would allow the vehicle to "remember" the driver’s habits over the long term, eliminating the need for re-learning at the start of every drive, thus providing a seamless personalized experience.

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